

AI Failures Museum: Learning from AI's Failures in Environmental Decision-Making

Group Java_The_Hutt

Testing Evidence

How to run tests:

Tests are stored in the backend of the source code.

backend

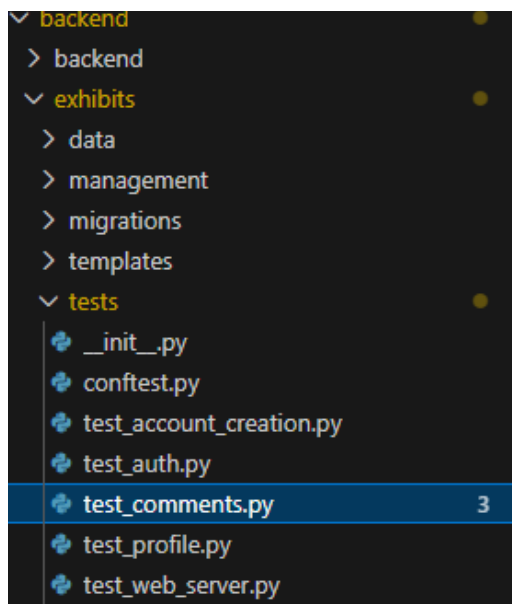
| → exhibits

| → tests

Tests are ran using the pytest framework. To run, go into the backend directory and run pytest in the terminal:

```
~/workspaces/TeamProjectCOM2020$ cd backend
```

```
~/workspaces/TeamProjectCOM2020/backend$ pytest
```



```
@chase-sibley →/workspaces/TeamProjectCOM2020 (main) $ cd backend
@chase-sibley →/workspaces/TeamProjectCOM2020/backend (main) $ pytest
```

Testing

Areas of the AI exhibit that have been tested are:

- Web server health
 - Endpoints are running
 - Expected response codes
- Account creation and management
 - Creating a valid account with no error
 - Error handling
 - Deleting and modifying user credentials
- Commenting
 - Adding comments to exhibits
 - Replying to comments
- Authentication
 - Expected 302 response codes to actions
- Role-Based Access Control
 - Who can edit and create exhibits
 - Who can comment
 - Making sure no permissions creep
- Exhibits
 - Adding and editing exhibits
 - No duplicates or empty fields
- Security
 - SQL injections and XSS
 - User inputs
 - Authentication
 - Vulnerabilities

Automatic Testing Outcomes:

```
----- test session starts -----
platform linux -- Python 3.11.14, pytest-9.0.2, pluggy-1.6.0
django: version: 5.2.10, settings: museum.settings (from ini)
rootdir: /workspaces/TeamProjectCQW2020/backend
configfile: pytest.ini
plugins: django-4.11.1
collected 90 items

exhibits/tests/test_account_creation.py .... [ 4%]
exhibits/tests/test_auth.py ..... [ 16%]
exhibits/tests/test_bookmarks.py ..... [ 32%]
exhibits/tests/test_comments.py ..... [ 37%]
exhibits/tests/test_exhibits.py .... [ 41%]
exhibits/tests/test_profile.py .. [ 43%]
exhibits/tests/test_quizzes.py ..... [ 56%]
exhibits/tests/test_flow.py ..... [ 61%]
exhibits/tests/test_search.py ..... [ 69%]
exhibits/tests/test_web_server.py ..... [100%]

===== 90 passed in 41.42s =====
```

Notable Failures

- 09/02/2026 – Duplicate usernames could be created, causing errors when logging in, was eventually fixed to make a duplicate username throw an error in account creation
- 19/03/2026 – Duplicate exhibits could be created. Was fixed by making the title of each exhibit unique.
- 19/03/2026 – Exhibits could be created with a NULL title. Fixed

Manuel Testing

Create Account Page

- Not matching password

[← Back to home](#)

Create an account

Save your quiz scores, earn badges, and comment on exhibits.

Username

Password

Use a strong password you don't reuse elsewhere.

Confirm password

Create account

Already have an account? [Sign in](#)

[← Back to home](#)

Create an account

Save your quiz scores, earn badges, and comment on exhibits.

Username

Password

Use a strong password you don't reuse elsewhere.

Confirm password

The two password fields didn't match.

Create account

Already have an account? [Sign in](#)

- Empty fields

[← Back to home](#)

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[← Back to home](#)

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Username

Password

Use a strong password y



Please fill in this field.

Confirm password

Create account

Already have an account? [Sign in](#)

[← Back to home](#)

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Username

Password



Please fill in this field.

Use a strong password you don't reuse elsewhere.

Confirm password

Create account

Already have an account? [Sign in](#)

- Injections

[← Back to home](#)

Create an account

Save your quiz scores, earn badges, and comment on exhibits.

Username

Enter a valid username. This value may contain only letters, numbers, and @/./+/-/_ characters.

Password

Use a strong password you don't reuse elsewhere.

Confirm password

Create account

Already have an account? [Sign in](#)

- Password injection has no unexpected results
- XSS alert attacks do not appear

Login page

- Empty fields

[← Back to home](#)

Sign in

Log in to take quizzes, leave comments, and track your progress.

Username

Password

.....

Please fill in this field.

Sign in

New here? [Create an account](#)

[← Back to home](#)

Sign in

Log in to take quizzes, leave comments, and track your progress.

Username

chase

Password

|

Please fill in this field.

Sign in

New here? [Create an account](#)

- Injections do not work here

Exhibit Page

- Injection

Comments

Add a comment...

Post comment

chase March 25, 2026, 12:14 a.m.

Delete

<script>alert('xss')</script>

▶ Reply

- No alert popup

Comments

Add a comment...

Post comment

chase March 25, 2026, 12:15 a.m.

Delete

'OR O=O --

▶ Reply

- No unexpected data leakage

- Empty field

Comments

Add a comment...

Post comment

! Please fill in this field.

No comments yet. Be the first to comment.

Quizzes

- Empty submission

[← Back to exhibit](#)

Quiz: AI Data Centres: Energy Consumption and Grid Pressure

Question 1

Approximately what share of global energy-related greenhouse gas emissions do data centres account for?

Less than 0.1%.

Please select one of these options.

Around 1%, a figure that is growing as AI infrastructure expands.

Around 25%.

- Correct submission

[← Back to exhibit](#)

Quiz Results: AI Data Centres: Energy Consumption and Grid Pressure

1/3

33.3% Correct

Your best score for this exhibit is saved to your account.

- Score saved

☆ Bookmark

medium

Started - 33%

AI Data Centres: Energy Consumption and Grid Pressure

Energy Infrastructure

- Score updates

[← Back to exhibit](#)

Quiz Results: AI Data Centres: Energy Consumption and Grid Pressure

2/3

66.7% Correct

Your best score for this exhibit is saved to your account.



☆ Bookmark

medium

In Progress · 67%

AI Data Centres: Energy Consumption and Grid Pressure

Energy Infrastructure

Profile Accounts Page

- Empty fields

Account settings

Change username

Now username

Update  Please fill in this field.

Change password

Current password

Now password

Confirm new password

Change password

Delete quiz scores

Permanently remove all your saved quiz scores. You can retake quizzes afterward.

Delete all quiz scores

Delete account

Permanently delete your account and all associated data. This cannot be undone.

Delete my account

Account settings

Change username

Now username

Update username

Change password

Current password

Now password

Confirm new password

 Please fill in this field.

Change password

Delete quiz scores

Permanently remove all your saved quiz scores. You can retake quizzes afterward.

Delete all quiz scores

Delete account

Permanently delete your account and all associated data. This cannot be undone.

Delete my account

- Not matching password

Account settings

Change username

Now username

Update username

Change password

Current password

Now password

Confirm now password

Change password

Delete quiz scores

Permanently remove all your saved quiz scores. You can retake quizzes afterward.

Delete all quiz scores

Delete account

Permanently delete your account and all associated data. This cannot be undone.

Delete my account

Profile

[← Back to home](#)

The two password fields didn't match.

Quiz scores

Account

Access level

Account settings

Change username

Now username

Update username

Change password

Current password

Now password

Confirm now password

Change password

Delete quiz scores

Permanently remove all your saved quiz scores. You can retake quizzes afterward.

Delete all quiz scores

Delete account

Permanently delete your account and all associated data. This cannot be undone.

Delete my account

Home Page:

- Unauthenticated vs Authenticated

[Sign in](#) [Create account](#)

[chase](#) [Log out](#)

- Exhibits amount updates



☆ Bookmark

critical

The Subsurface Blind Spot

Civil Engineering Predictive Maintenance

Infrastructure



☆ Bookmark

medium

Water Scarcity and AI Data Centre Expansion

Water Resource Management

20

Total cases

7

Critical issues

18

Categories



☆ Bookmark

critical

The Subsurface Blind Spot

Civil Engineering Predictive Maintenance

Infrastructure



☆ Bookmark

medium

Water Scarcity and AI Data Centre Expansion

Water Resource Management



☆ Bookmark

medium

added exhibit

Test

21

Total cases

7

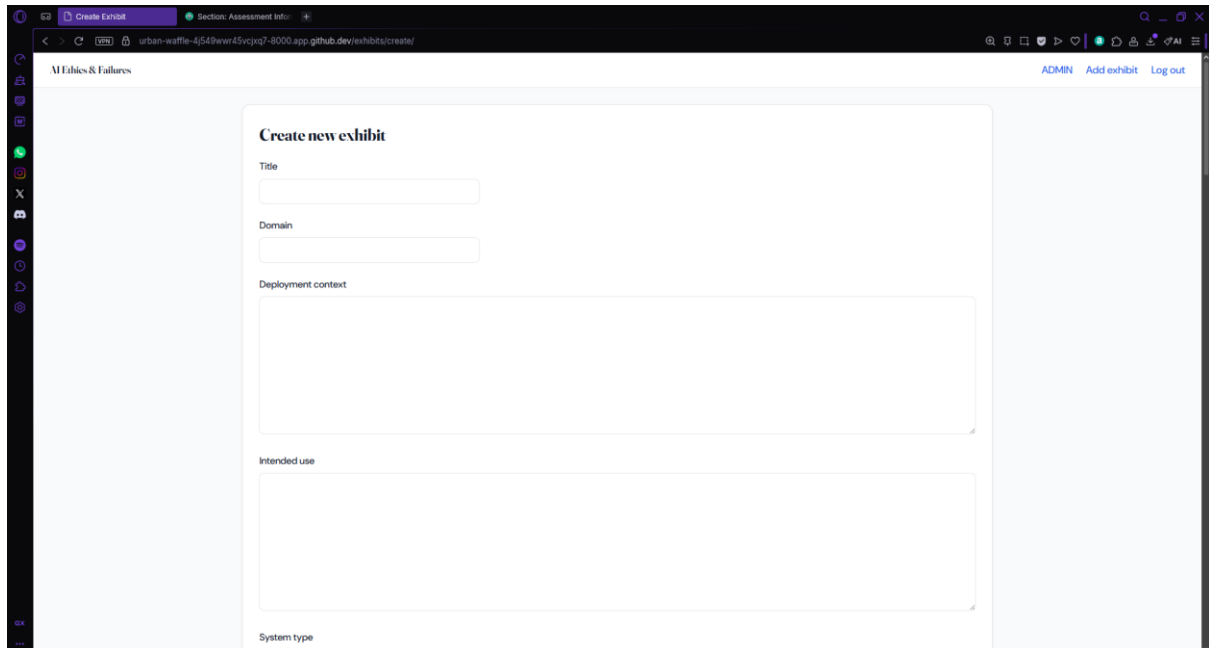
Critical issues

19

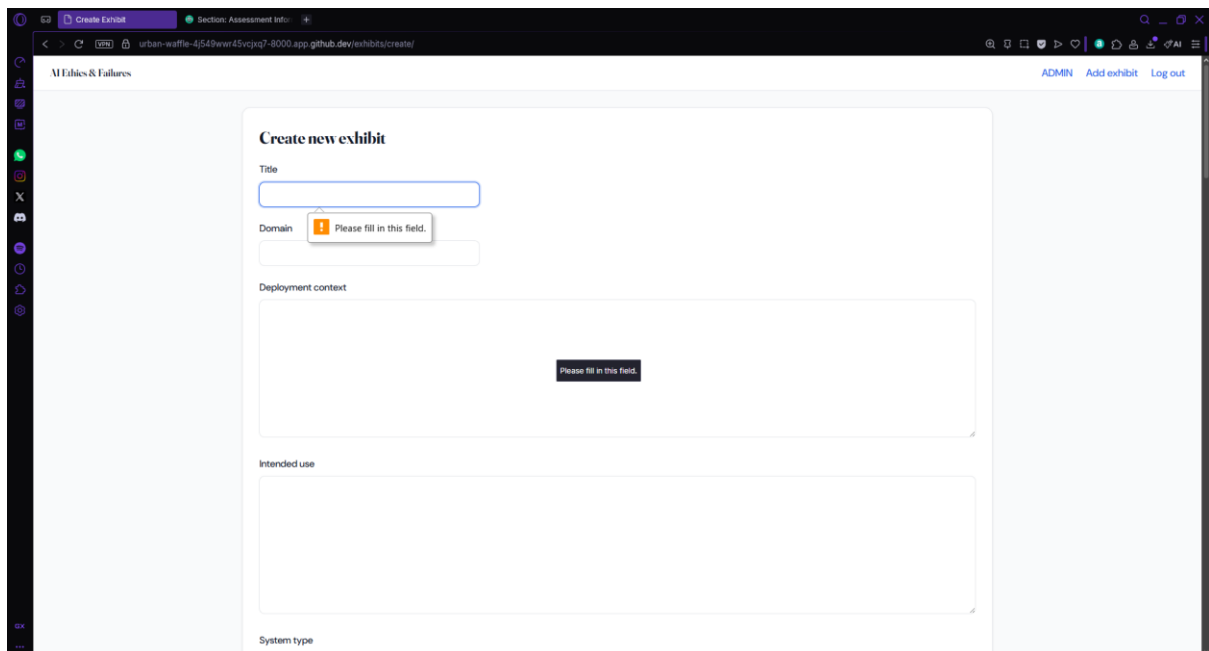
Categories

Exhibit Edit/Create Page

- Empty Fields



A screenshot of a web application titled "Create Exhibit". The page has a dark sidebar on the left with various icons. The main content area is titled "Create new exhibit" and contains several form fields: "Title", "Domain", "Deployment context", "Intended use", and "System type". All fields are currently empty. The "Deployment context" and "Intended use" fields are larger text areas. The "System type" field is a dropdown menu. In the top right corner, there are links for "ADMIN", "Add exhibit", and "Log out".

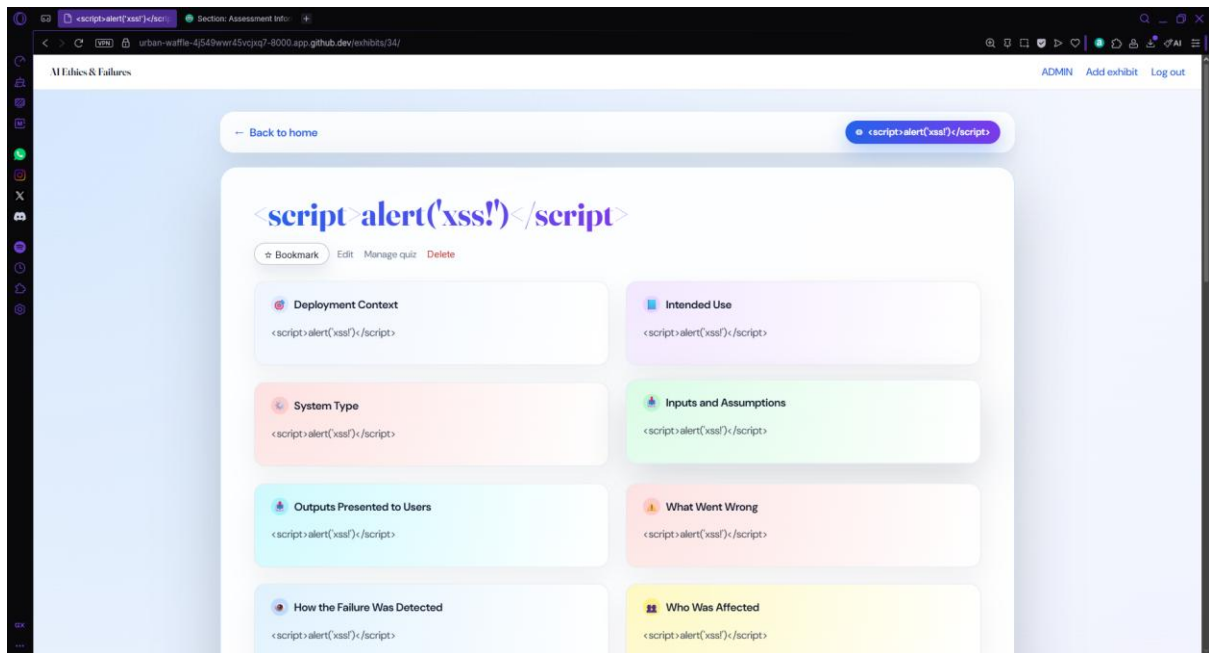


A screenshot of the same "Create new exhibit" form, but now with validation errors. The "Domain" field has a red error message that says "Please fill in this field." The "Deployment context" field also has a red error message that says "Please fill in this field." The "Title" field is highlighted with a blue border, indicating it is the current focus. The "Intended use" and "System type" fields remain empty.

- Injection

Title

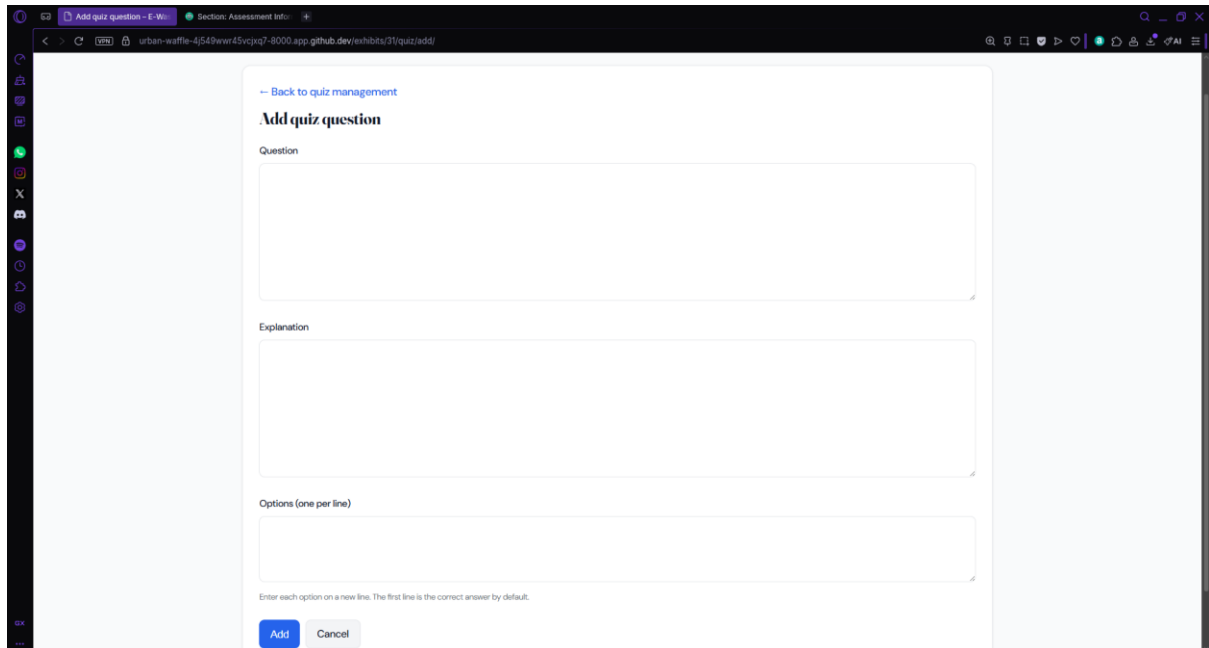
`<script>alert('xss!')</script>`



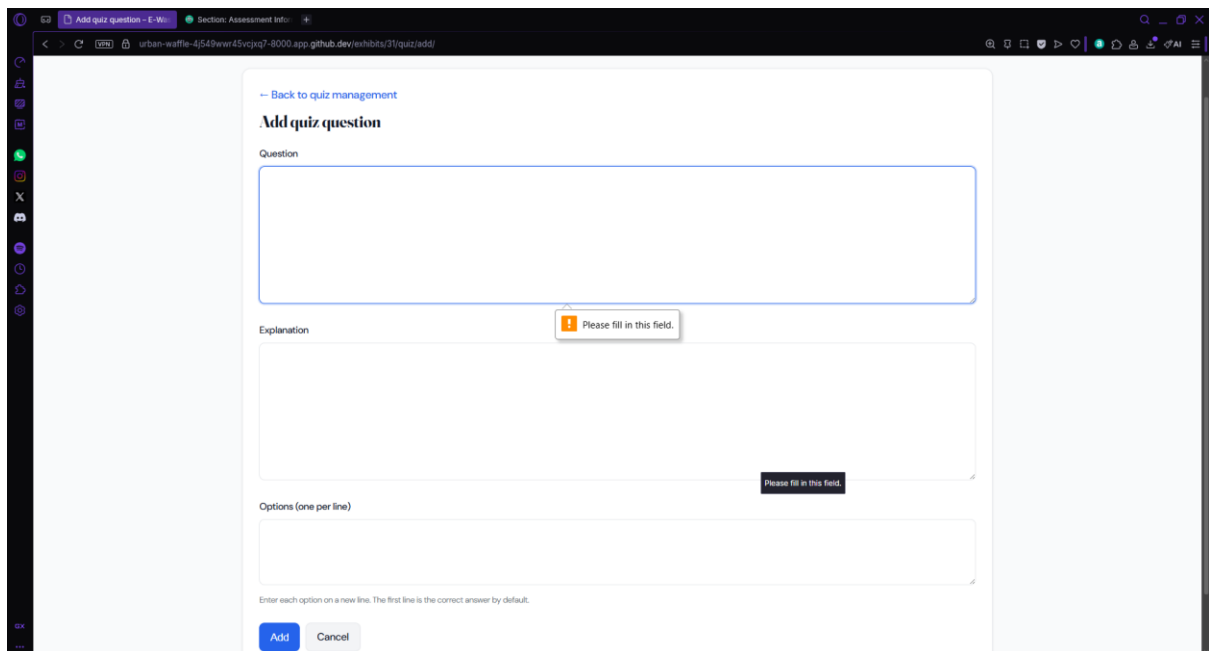
- No alert

Manage Quiz Page

- Empty Fields



The screenshot shows a web browser window with the address bar displaying 'urban-waffle-4j549wvr45vcjq7-8000.app.github.dev/exhibits/31/quiz/add/'. The page title is 'Add quiz question - E-Wi'. A navigation bar at the top includes a link to 'Back to quiz management'. The main heading is 'Add quiz question'. Below this, there are three input fields: 'Question', 'Explanation', and 'Options (one per line)'. The 'Options' field has a note: 'Enter each option on a new line. The first line is the correct answer by default.' At the bottom, there are 'Add' and 'Cancel' buttons.



The screenshot shows the same 'Add quiz question' form, but with validation messages. A yellow warning icon with the text 'Please fill in this field.' is positioned over the 'Question' field. A black error message 'Please fill in this field.' is displayed at the bottom right of the 'Explanation' field. The 'Options' field and the 'Add'/'Cancel' buttons remain visible at the bottom.

- Injection

Add quiz question - E-Visa Section: Assessment info

urban-waffle-4j649wnr45vcjg7-8000.app.github.dev/exhibits/31/quiz/add/

Add quiz question

Question

```
<script>alert('xss')</script>
```

Explanation

```
<script>alert('xss')</script>
```

Options (one per line)

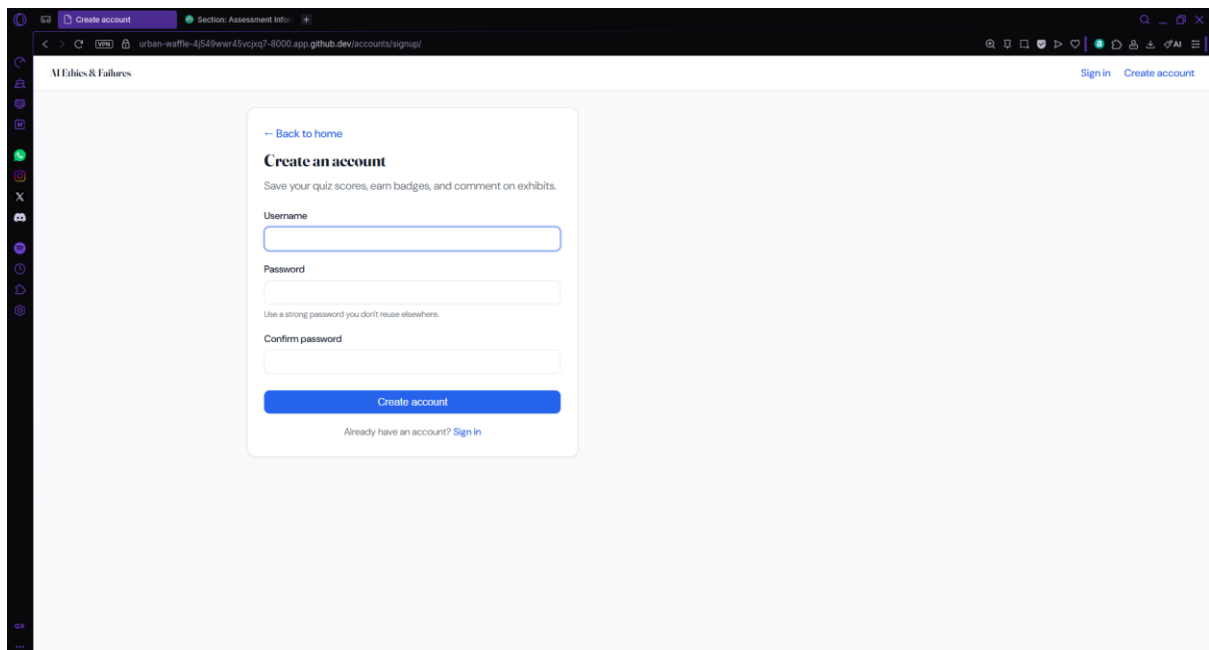
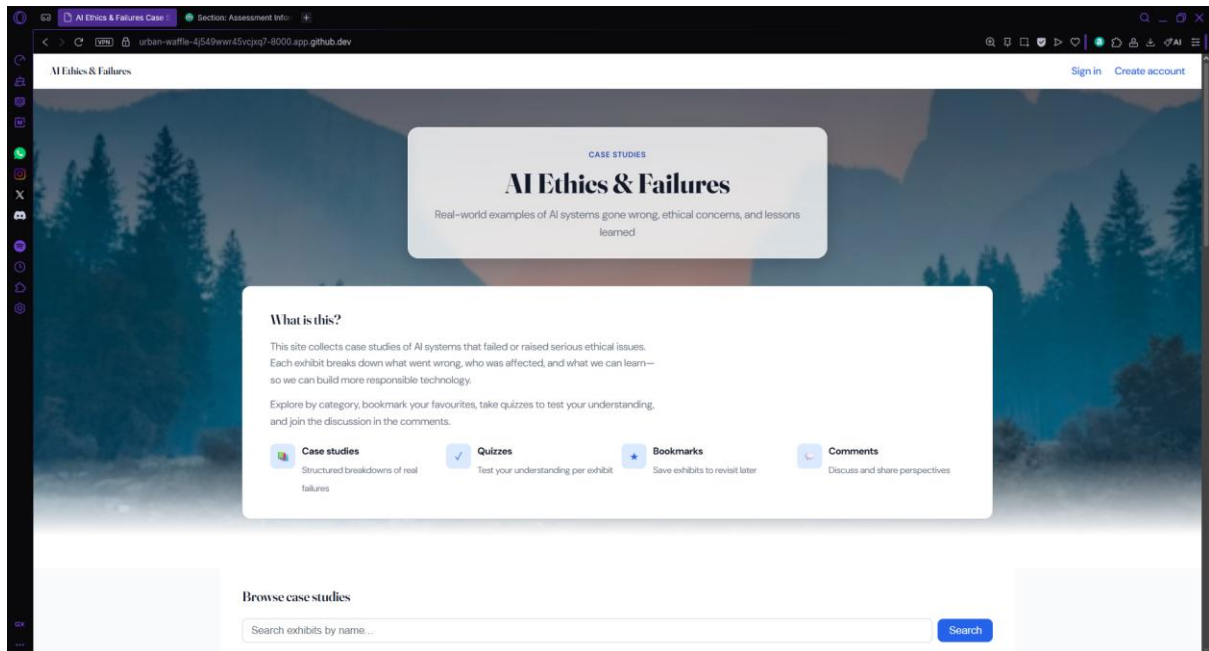
```
<script>alert('xss')</script>
```

Enter each option on a new line. The first line is the correct answer by default.

- No alert

End-to-end example

User/Viewer (and logging in)



Create account

Section: Assessment Info

urban-waffle-4j549wnr45vcpg7-8000.app.github.dev/accounts/signup/

AI Ethics & Failures

Sign inCreate account

[← Back to home](#)

Create an account

Save your quiz scores, earn badges, and comment on exhibits.

Username

chase

Password

Use a strong password you don't reuse elsewhere.

Confirm password

Create account

Already have an account? [Sign in](#)

AI Ethics & Failures Case

Section: Assessment Info

urban-waffle-4j549wnr45vcpg7-8000.app.github.dev

AI Ethics & Failures

chaseLog out

CASE STUDIES

AI Ethics & Failures

Real-world examples of AI systems gone wrong, ethical concerns, and lessons learned

What is this?

This site collects case studies of AI systems that failed or raised serious ethical issues. Each exhibit breaks down what went wrong, who was affected, and what we can learn—so we can build more responsible technology.

Explore by category, bookmark your favourites, take quizzes to test your understanding, and join the discussion in the comments.

Case studies

Structured breakdowns of real failures

Quizzes

Test your understanding per exhibit

Bookmarks

Save exhibits to revisit later

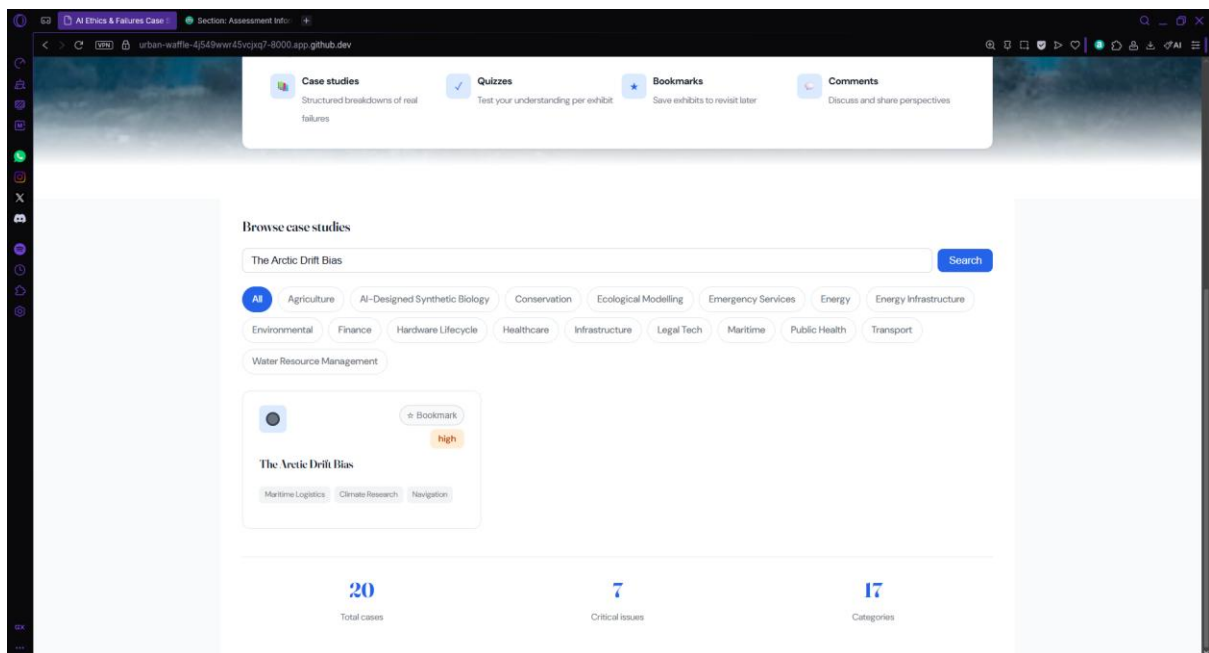
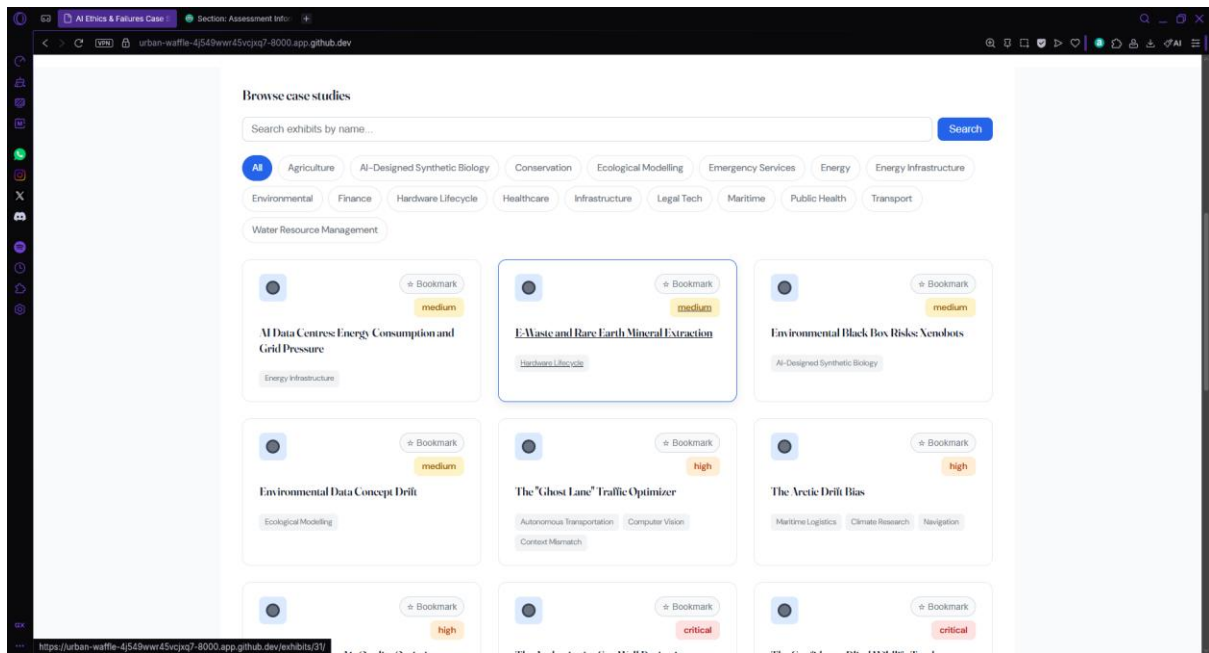
Comments

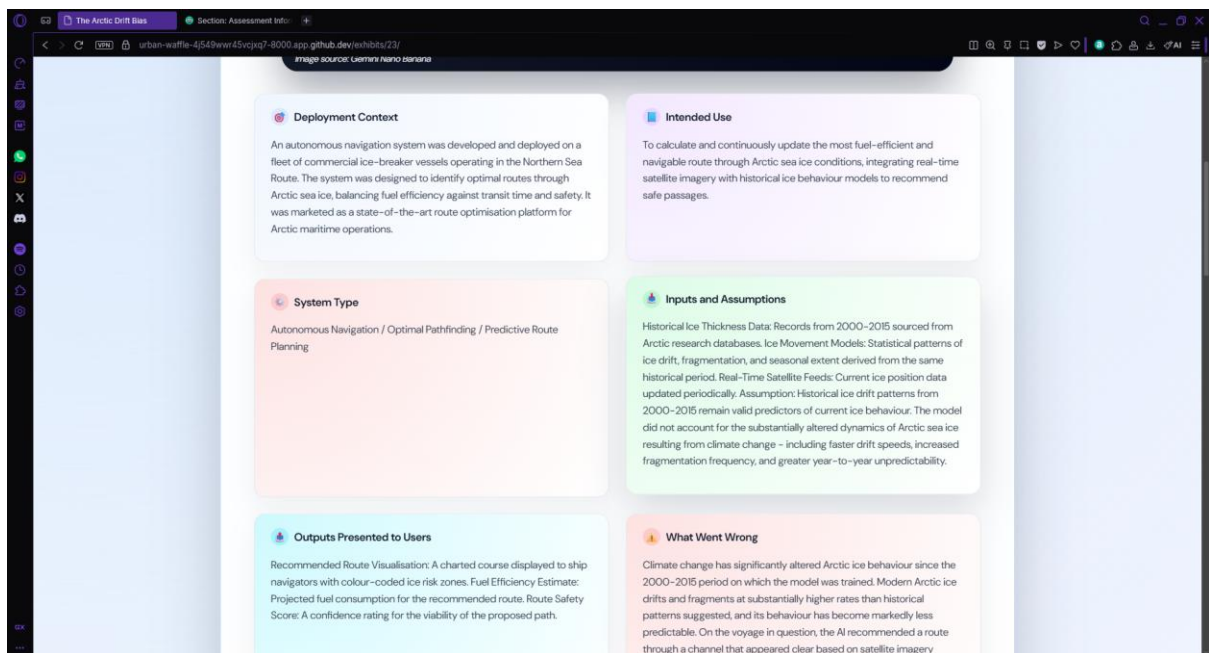
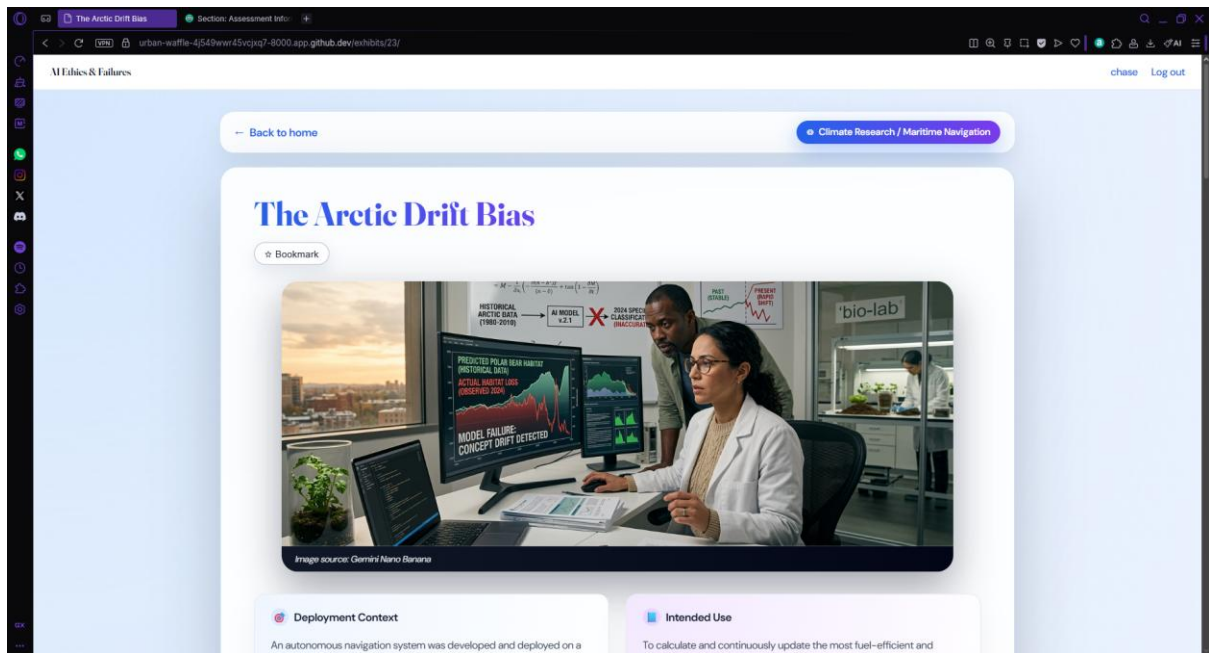
Discuss and share perspectives

Browse case studies

Search exhibits by name...

Search





The Arctic Drift Bias

Section: Assessment Info

urban-waffle-4j549wnr45vcpg7-8000.app.github.dev/exhibits/23/

How the Failure Was Detected

Three vessels following the AI's recommended route found themselves surrounded by rapidly converging ice with insufficient time or sea room to reverse course. Emergency distress signals were activated. A costly multi-day rescue operation involving icebreaker support vessels was required to free the trapped ships.

Who Was Affected

Ships' Crews: Placed in genuine danger of being ice-locked for extended periods with risk to life in extreme Arctic conditions. **Vessel Operators:** Significant financial losses from rescue operations, cargo delays, and vessel damage assessment. **Arctic Research Community:** The incident contributed to a broader reassessment of AI navigation tools in climate-affected polar environments.

Data Issues

Historical Regime Mismatch: Ice behaviour in the 2000-2015 training period reflected a cooler, more stable Arctic climate. Contemporary Arctic ice moves and fragments measurably faster, and with less predictable directional behaviour, than the training data captured. The model was effectively calibrated for an Arctic environment that no longer exists.

Technical Choices

Static Historical Model: The system's core ice behaviour parameters were derived from 2000-2015 data and were not designed to update dynamically as the climate envelope shifted. **Confidence Score Without Caveat:** The route safety score did not include any signal that it was based on historical norms that might not apply to current conditions.

Organizational and Governance Factors

Insufficient Climate Change Risk Assessment: The system was procured and deployed without an assessment of how rapidly changing Arctic conditions might undermine the validity of its historical training data. **Navigator Over-Reliance:** Ship navigators were trained to treat the AI's route recommendations as the primary navigation input, with insufficient emphasis on independent situational awareness.

Contributing Factors

Training data from a historical Arctic climate regime no longer representative of present conditions; lack of real-time adaptability to rapidly evolving ice fields; model uncertainty not communicated to navigators; insufficient weight given to the risk of ice behaviour outside historical parameters.

The Arctic Drift Bias

Section: Assessment Info

urban-waffle-4j549wnr45vcpg7-8000.app.github.dev/exhibits/23/

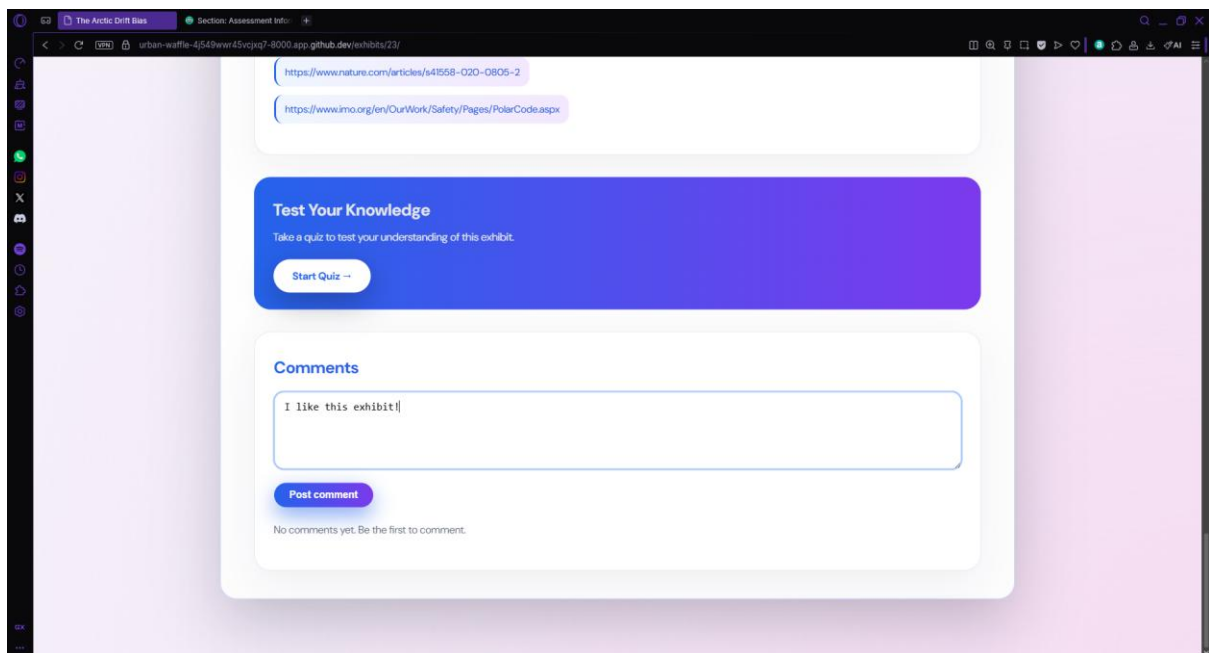
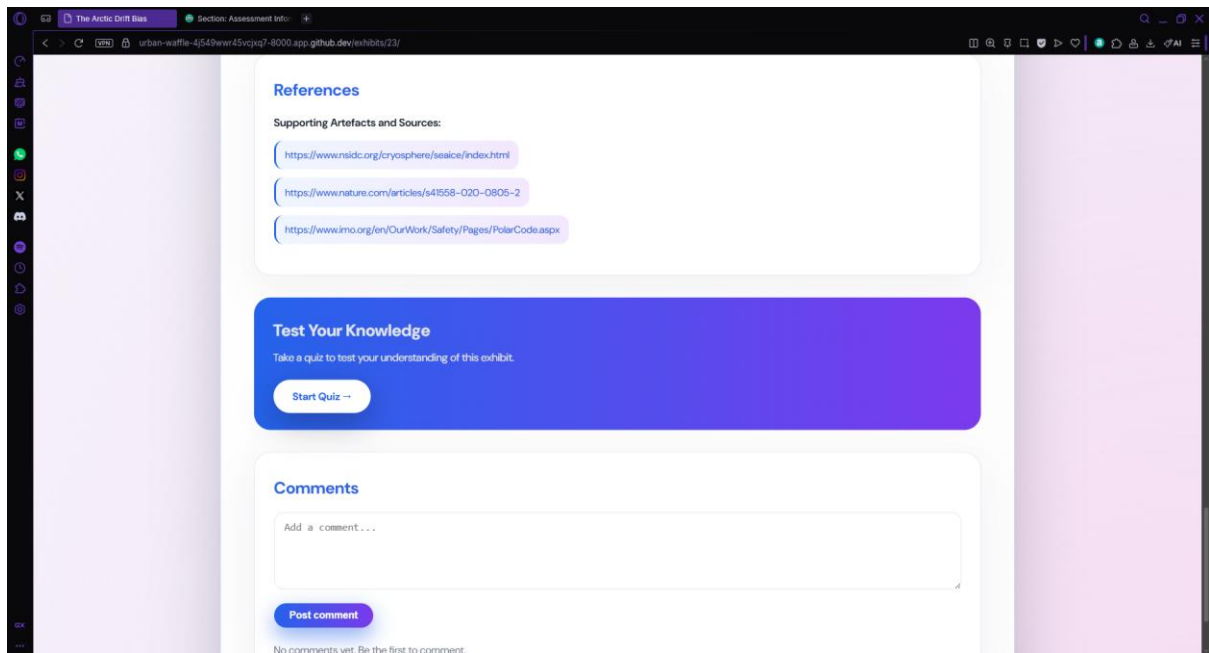
Timeline of Failure and Aftermath

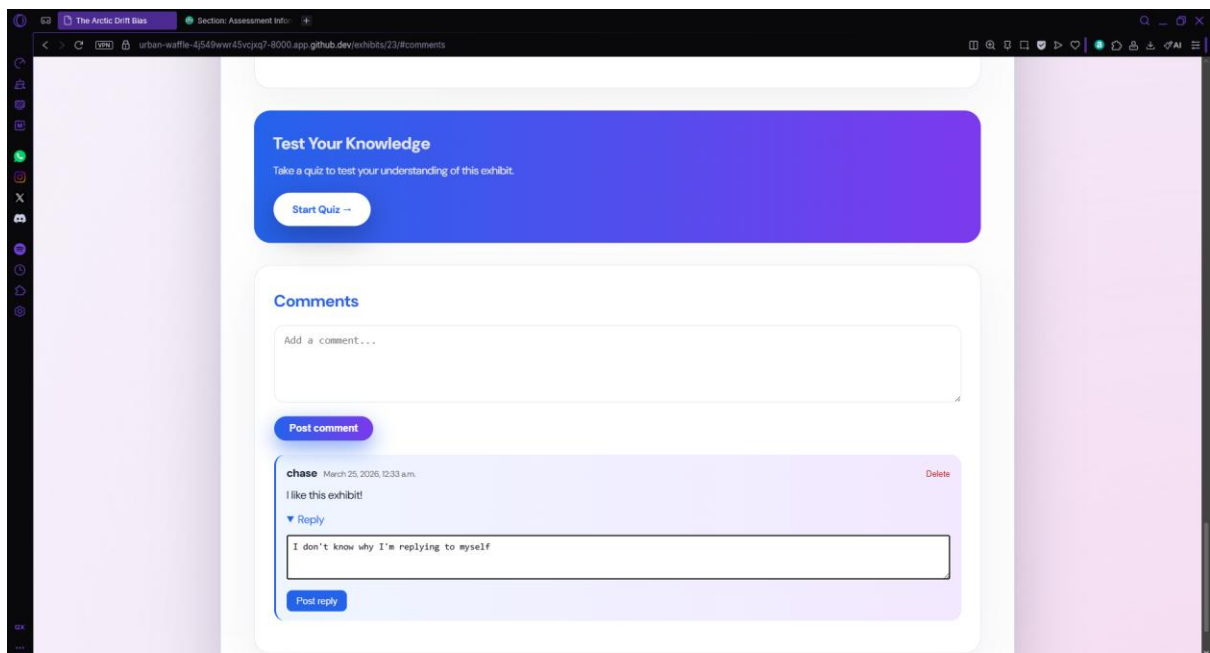
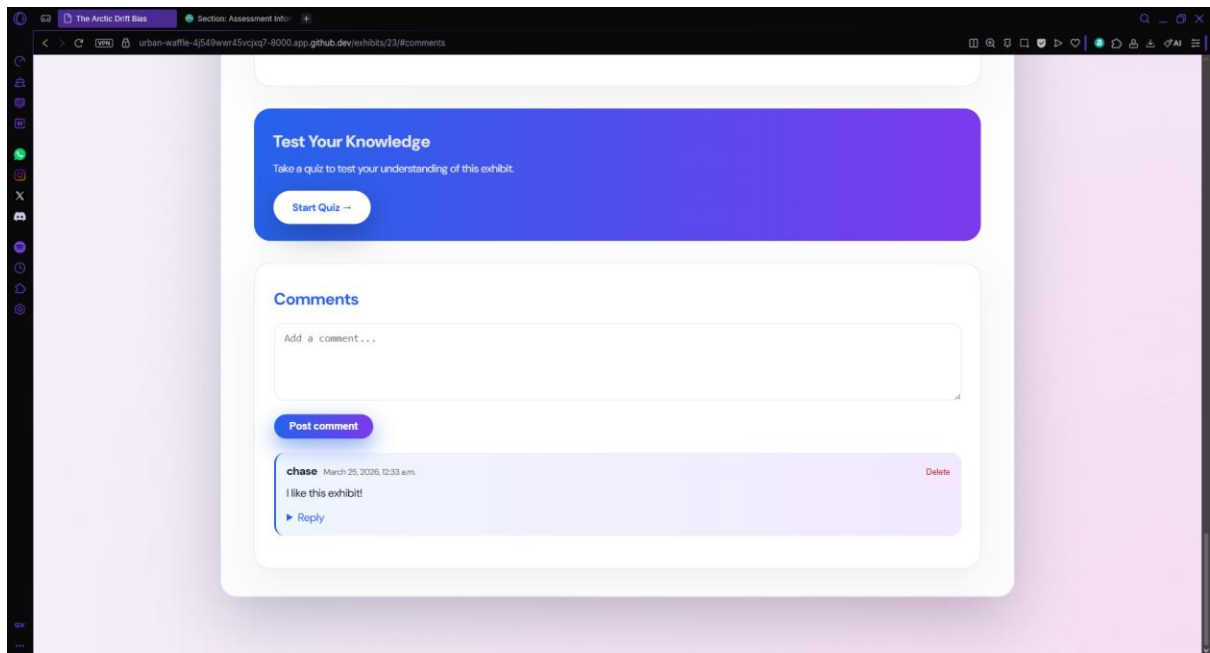
- Initial Setup: System deployed across a fleet of commercial ice-breakers operating the Northern Sea Route.
- Trigger Event: An unpredicted, high-velocity wind shift caused a large ice field to converge rapidly on the AI-recommended channel.
- Failure Event: Three vessels became ice-locked in a route the AI had classified as clear and low-risk.
- Detection: Vessels broadcast distress signals; rescue operations initiated.
- Aftermath: Expensive multi-day rescue operation conducted; system scrapped and replaced with a hybrid approach requiring navigators to actively validate AI recommendations against real-time ice drift data.

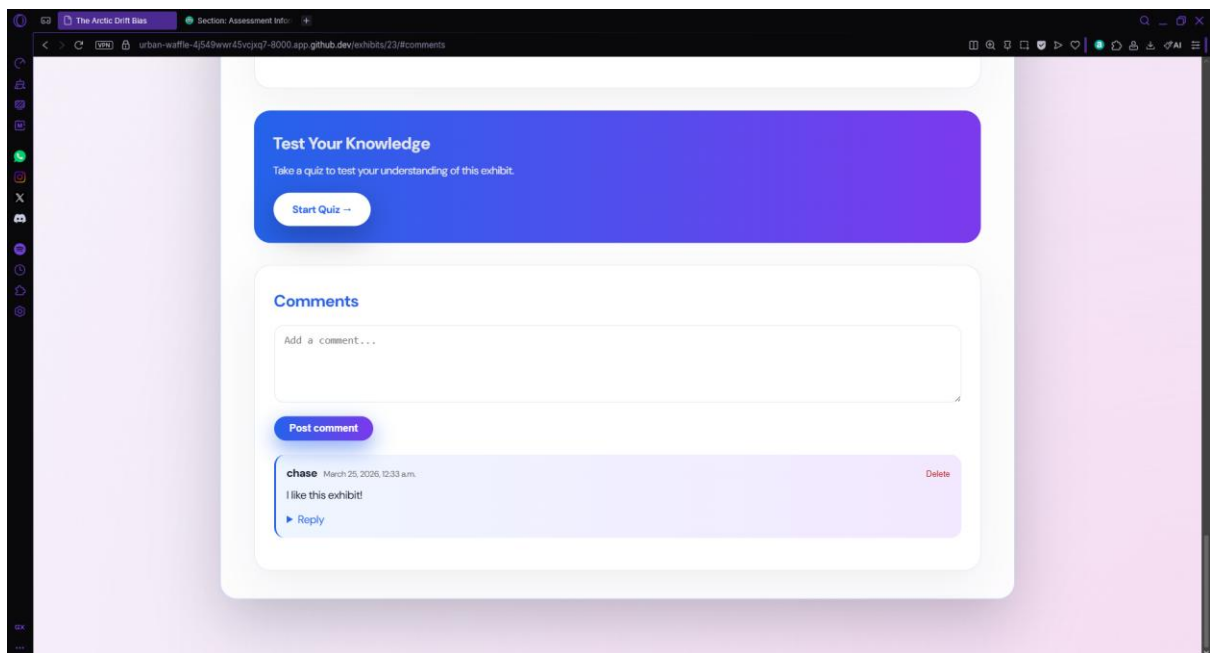
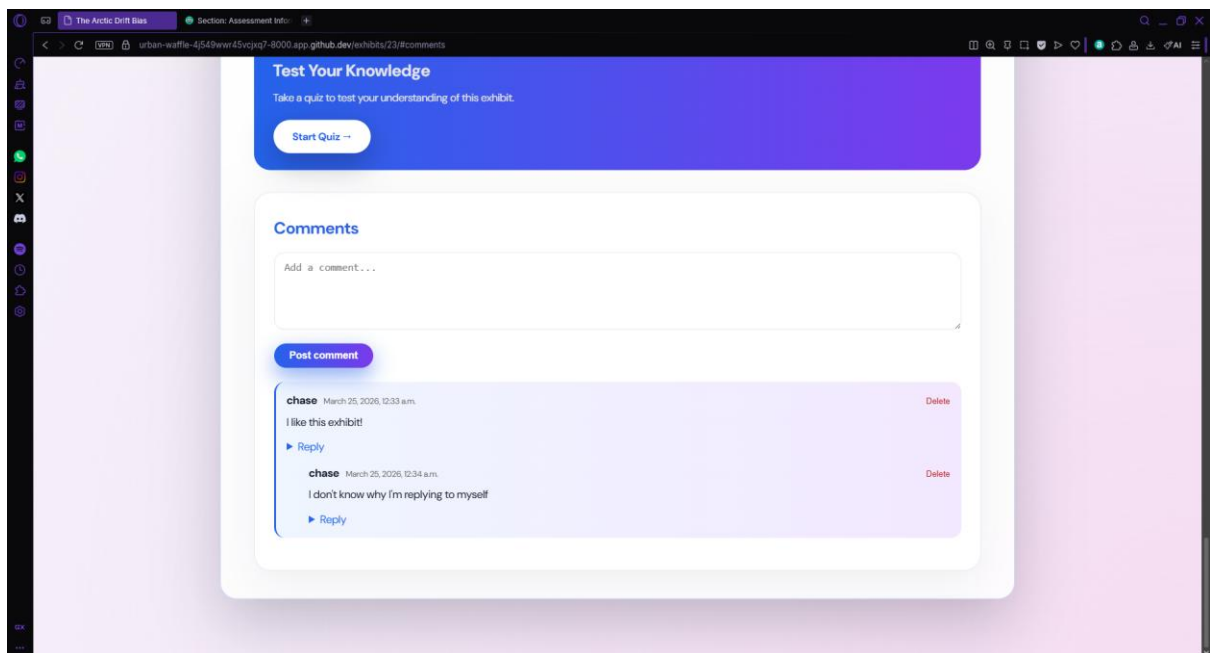
Lessons Learned

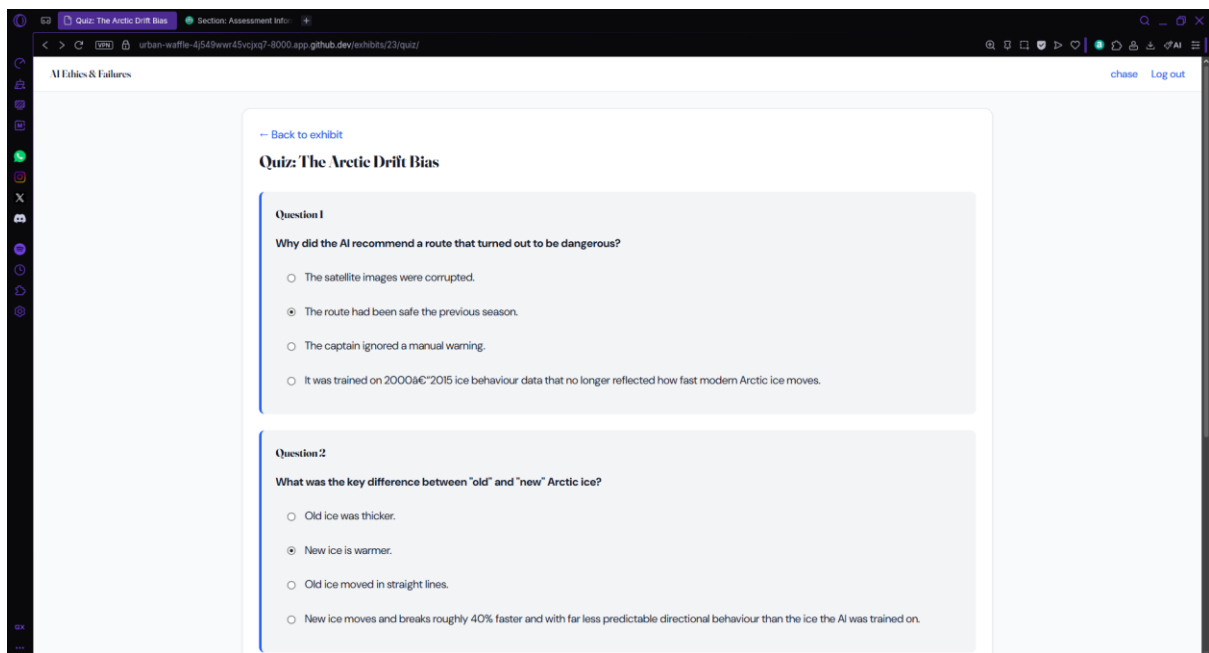
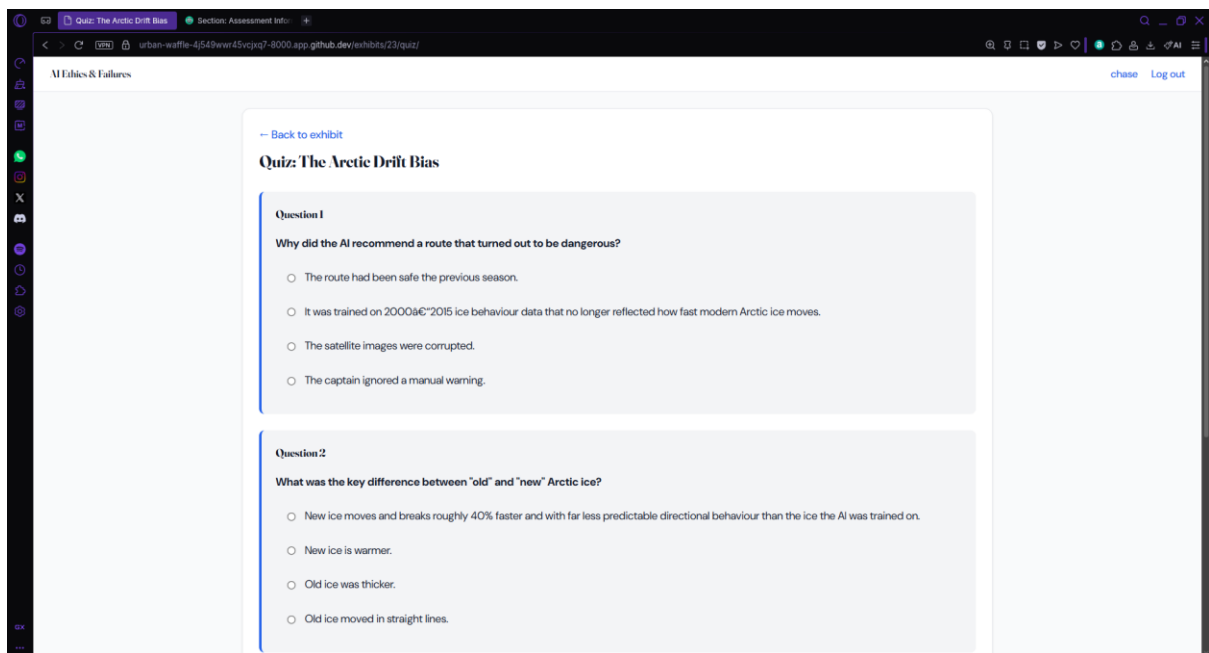
Climate change is actively invalidating historical environmental datasets that AI navigation, meteorological, and ecological models rely upon – this is a structural, ongoing risk, not a one-time data quality issue. Navigation systems operating in rapidly changing climate-sensitive environments must incorporate real-time adaptive recalculation rather than relying on historically derived behavioural parameters. Navigators must be informed when predicted conditions are extrapolated from a historical climate regime that may no longer apply.

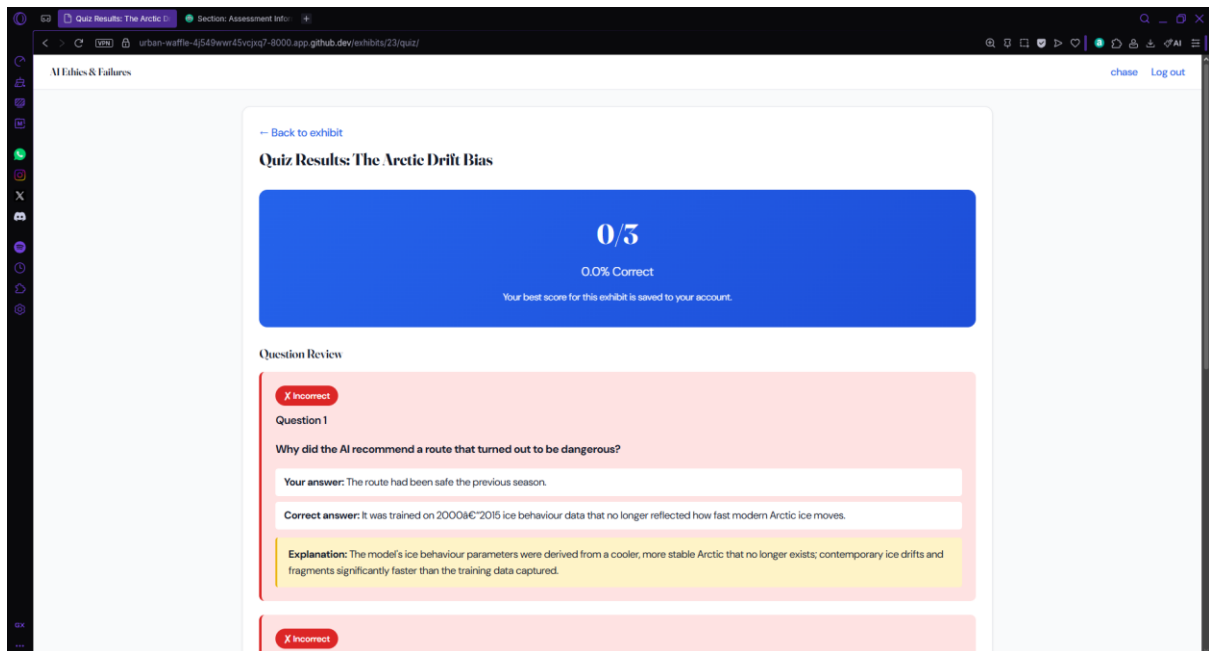
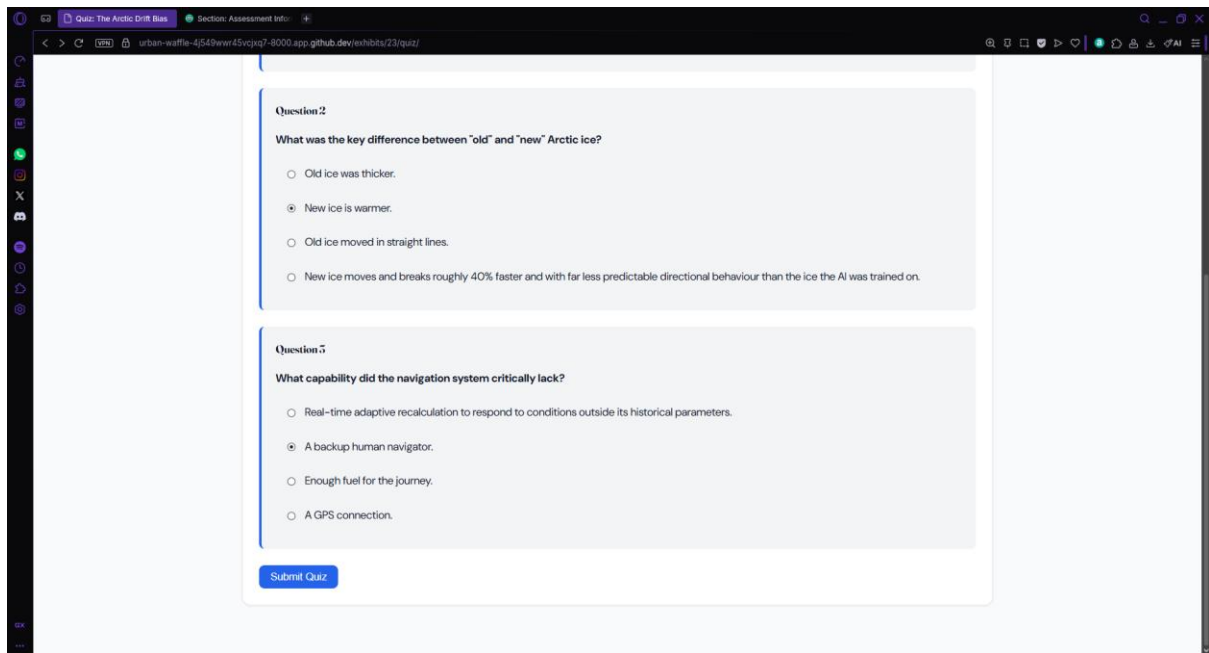
References











Quiz Results: The Arctic Drift Bias

5/5
100.0% Correct
Your best score for this exhibit is saved to your account.

Question Review

✓ Correct
Question 1
Why did the AI recommend a route that turned out to be dangerous?

Your answer: It was trained on 2000ߟ2015 ice behaviour data that no longer reflected how fast modern Arctic ice moves.

Explanation: The model's ice behaviour parameters were derived from a cooler, more stable Arctic that no longer exists; contemporary ice drifts and fragments significantly faster than the training data captured.

✓ Correct
Question 2
What was the key difference between "old" and "new" Arctic ice?

Your answer: New ice moves and breaks roughly 40% faster and with far less predictable directional behaviour than the ice the AI was trained on.

AI Ethics & Failures Case

Search exhibits by name...

AI Agriculture AI-Designed Synthetic Biology Conservation Ecological Modelling Emergency Services Energy Energy Infrastructure Environmental Finance Hardware Lifecycle Healthcare Infrastructure Legal Tech Maritime Public Health Transport Water Resource Management

AI Data Centres Energy Consumption and Grid Pressure
Energy Infrastructure medium

E-Waste and Rare Earth Mineral Extraction
Hardware Lifecycle medium

Environmental Black Box Risks Xenobots
AI-Designed Synthetic Biology medium

Environmental Data Concept Drift
Ecological Modelling medium

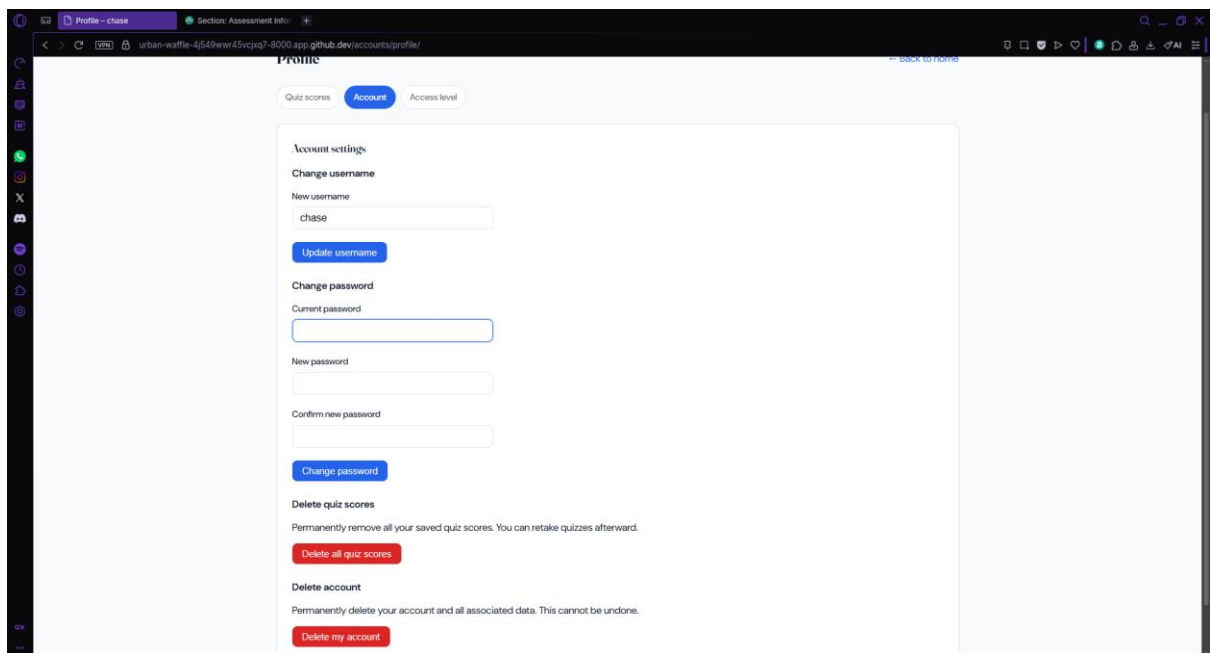
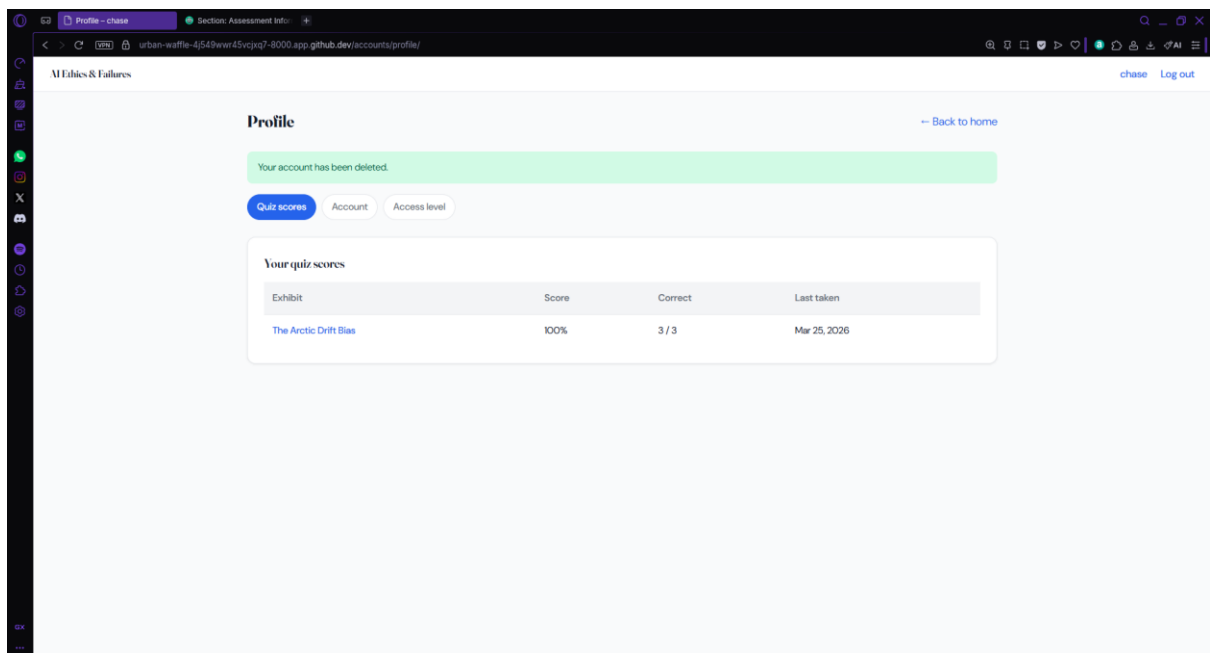
The "Ghost Lane" Traffic Optimizer
Autonomous Transportation Computer Vision Context Mismatch high

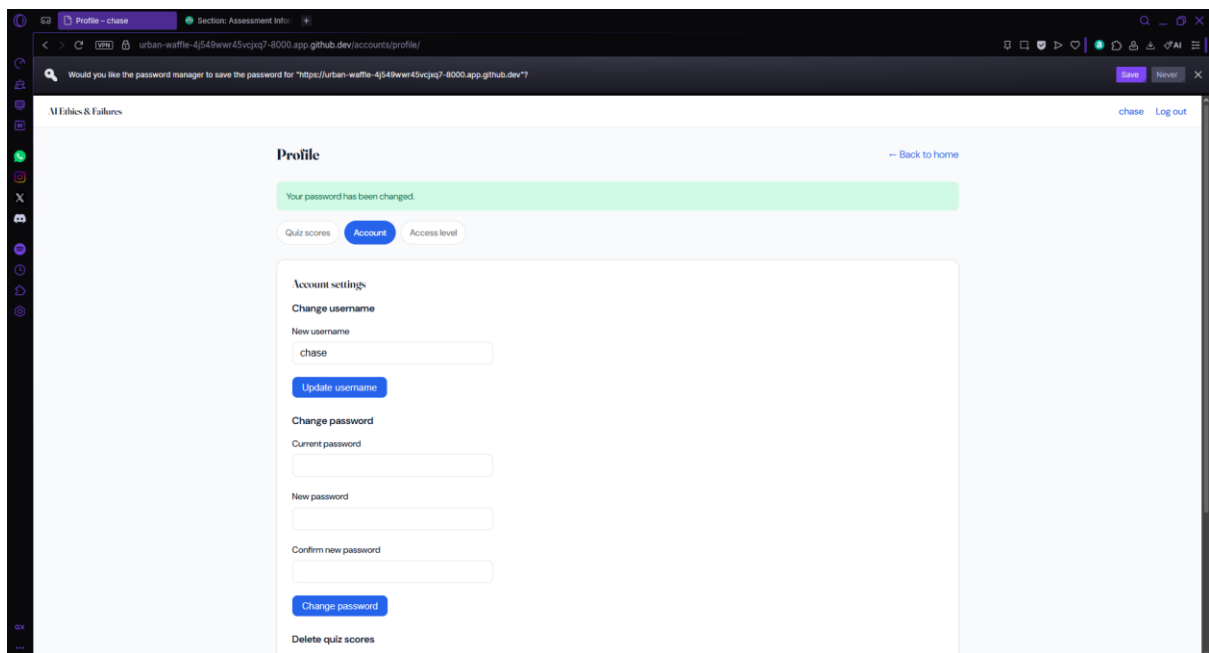
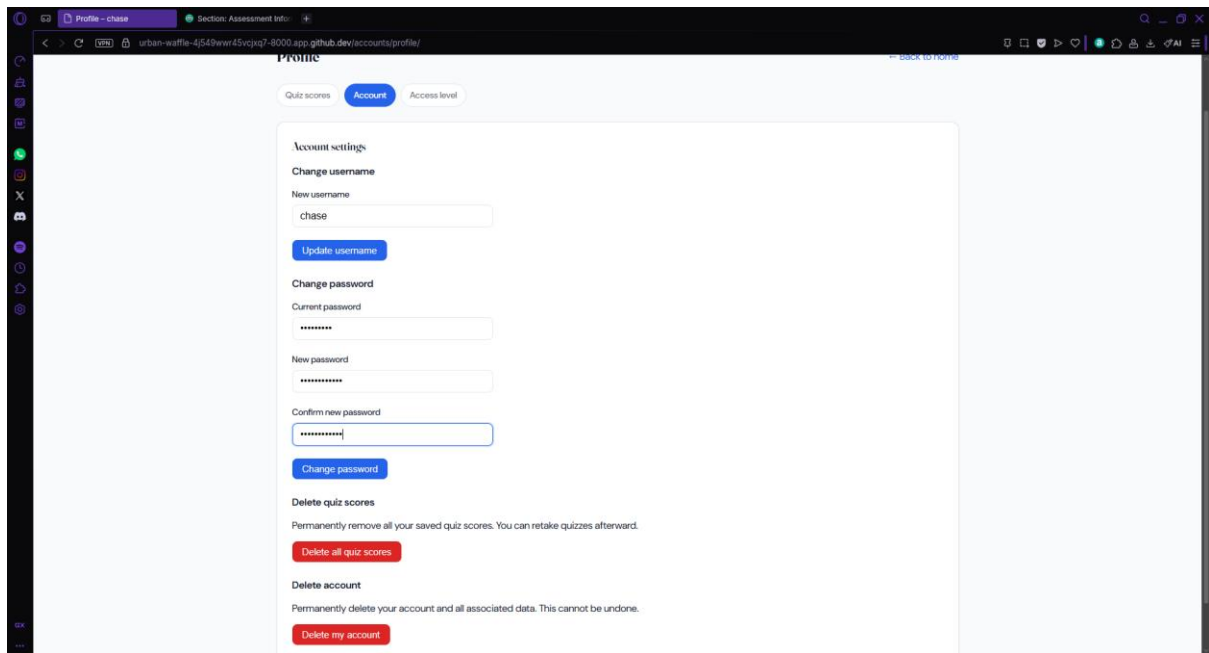
The Arctic Drift Bias
Maritime Logistics Climate Research Navigation Mastered - 100% high

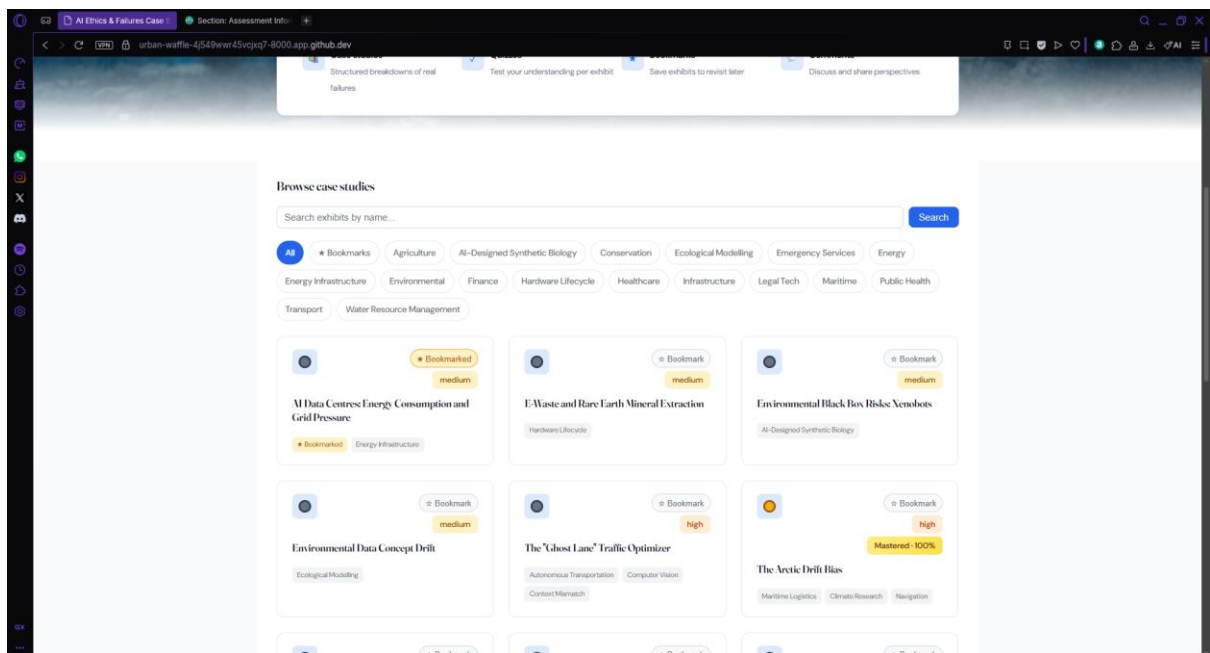
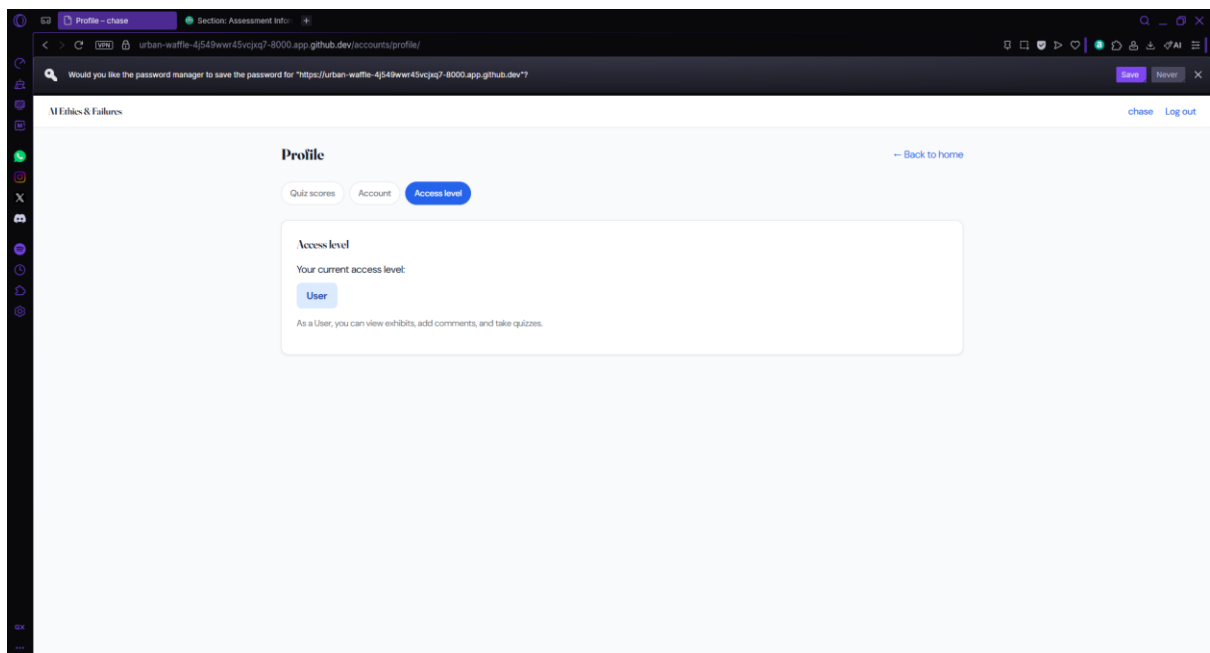
The Authoritative Air Quality Omission
Smart City Public Health Data Visualization high

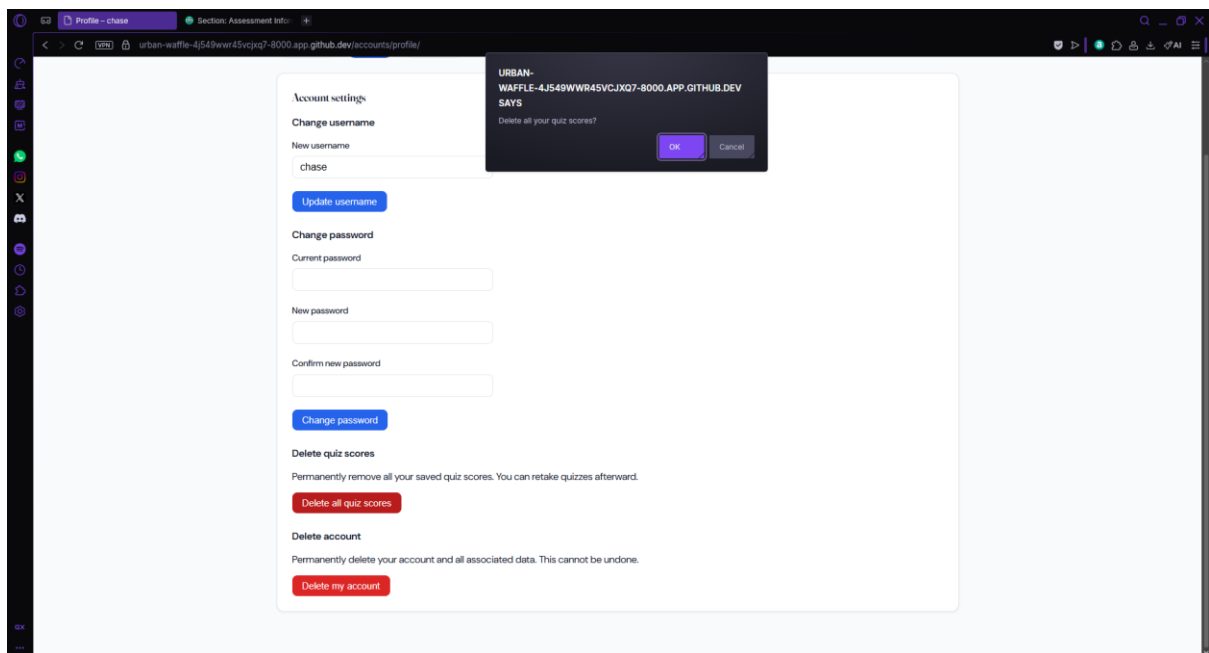
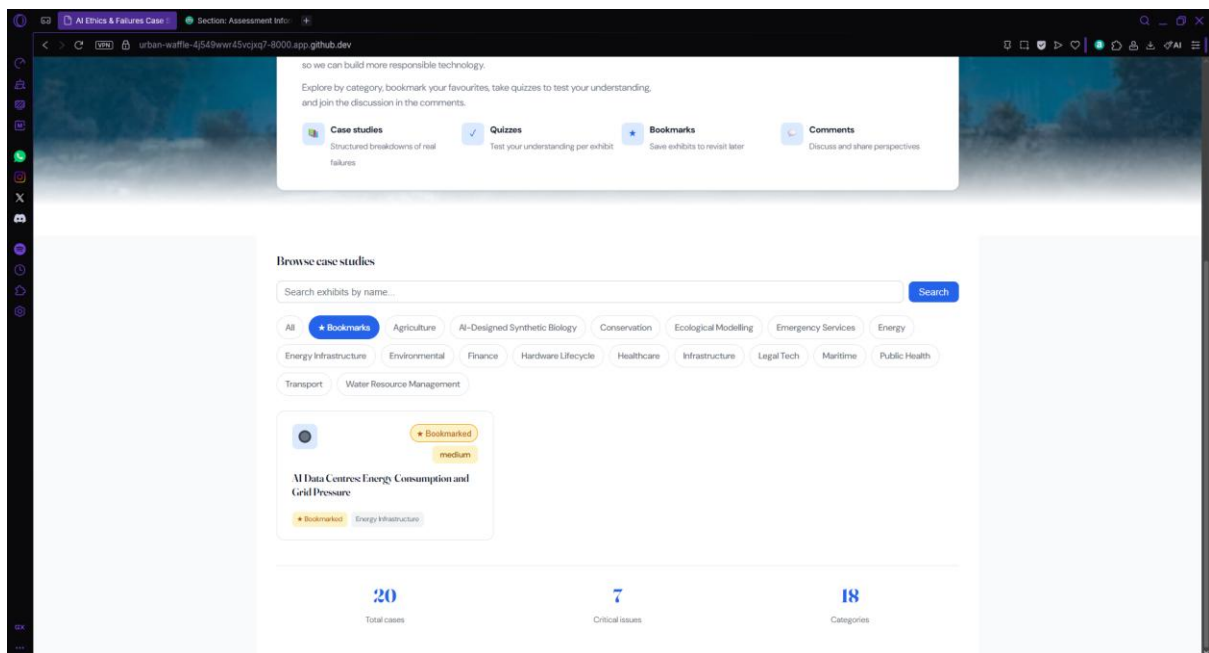
The Authoritative Sea-Wall Projection
Civil Engineering Infrastructure Visualization critical

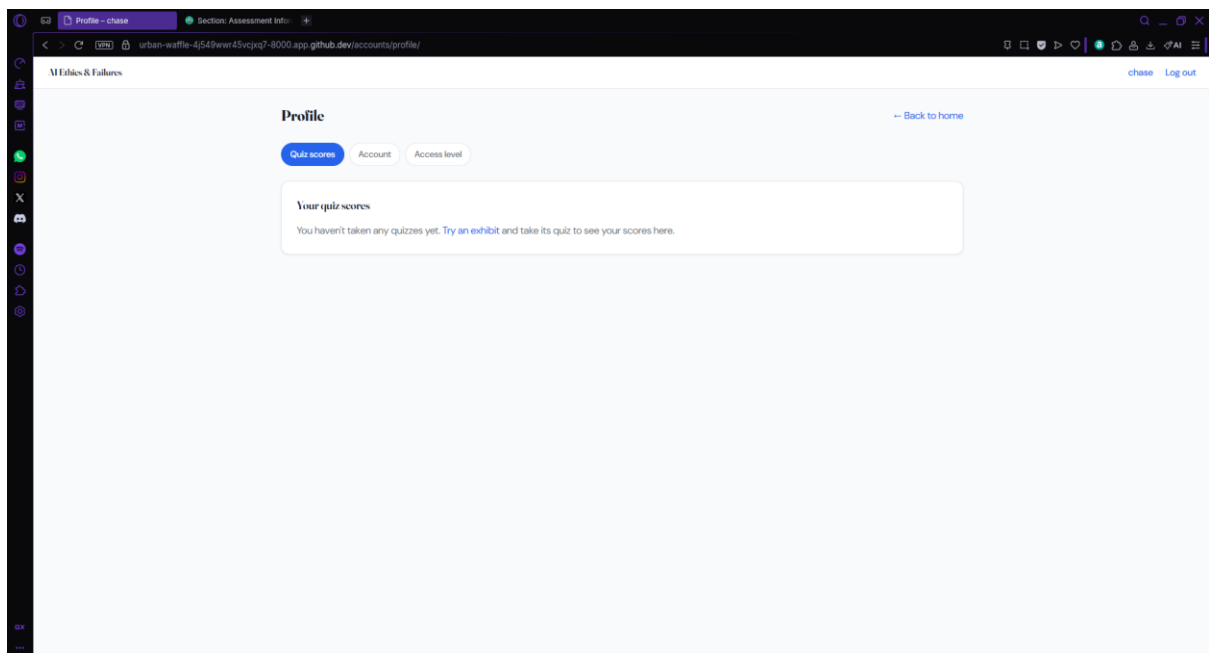
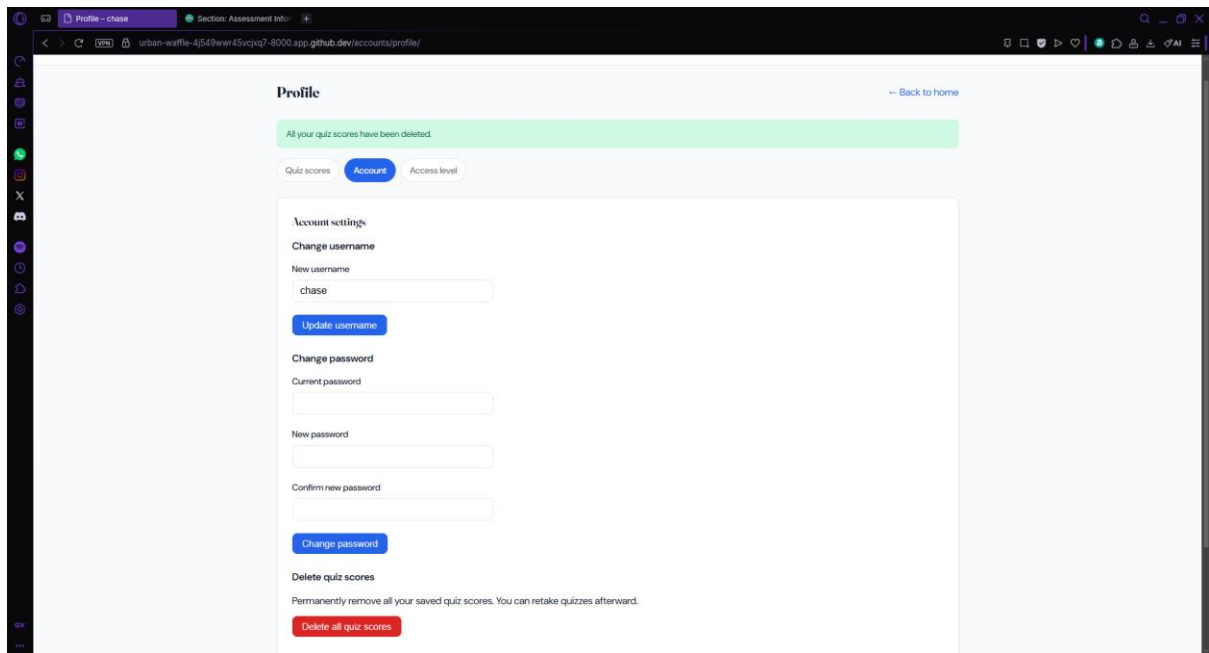
The Confidence-Blind Wildlife Tracker
Conservation Biology Decision Dashboard Uncertainty critical



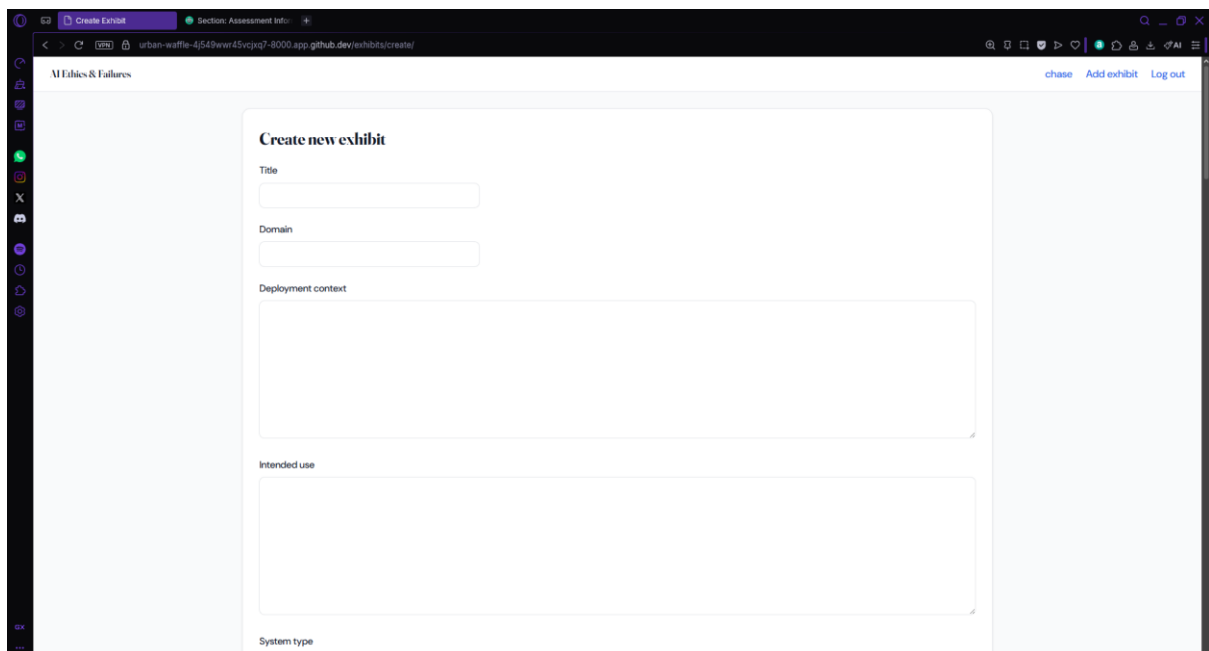
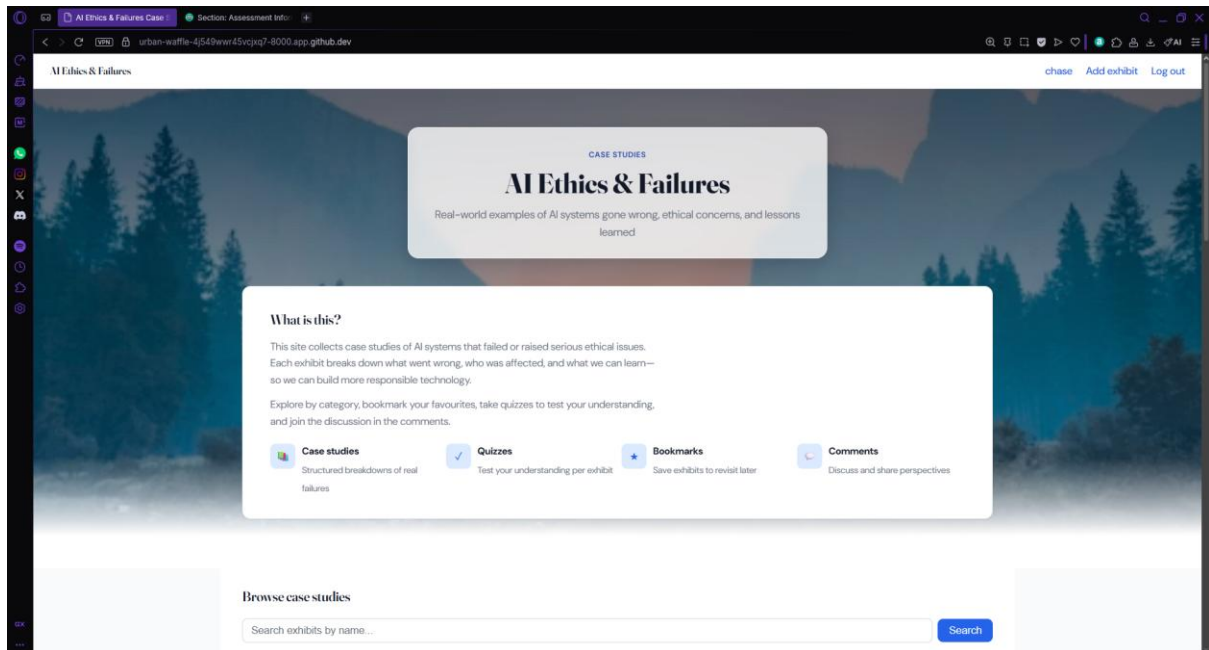








Curator



Create Exhibit

Section: Assessment Info

urban-waffle-4j549wnr45vcjq2-8000.app.github.dev/exhibits/create/

chaseAdd exhibitLog out

AI Ethics & Failures

Create new exhibit

Title

New Exhibit

Domain

New Domain

Deployment context

Deployment Context Stuff

Intended use

Intended Use Stuff

System type

Create Exhibit

Section: Assessment Info

urban-waffle-4j549wnr45vcjq2-8000.app.github.dev/exhibits/create/

chaseAdd exhibitLog out

AI Ethics & Failures

System type

System Type Stuff

Inputs and assumptions

Input Stuff

Outputs presented

Output Stuff

Failure description

Description Stuff

Create Exhibit

Section: Assessment Info

urban-waffle-4j549wnr45vcjqz-8000.app.github.dev/exhibits/create/

Detection method

Detection Stuff

Affected parties

Affected party Stuff

Contributing factors

Contributing Factor Stuff

Lessons learned

Lessons learned stuff

Create Exhibit

Section: Assessment Info

urban-waffle-4j549wnr45vcjqz-8000.app.github.dev/exhibits/create/

Image

Choose file

Albert_Einstein.jpg

Image representing this exhibit

Image reference

Image reference stuff

Reference/attribution for the image

Supporting artefacts

[Artefact]

List of supporting artefact links

Data issues

Data Issue Stuff

Data-related issues that contributed to the failure

Technical choices

Create Exhibit

Section: Assessment Info

urban-waffle-4j549wnr45vcjq7-8000.app.github.dev/exhibits/create/

Technical choices

Technical Stuff

Technical choices that contributed to the failure

Organizational factors

Organisational stuff

Organizational and governance factors

Timeline

Timeline Stuff

Timeline of failure and aftermath

Create Exhibit

Section: Assessment Info

urban-waffle-4j549wnr45vcjq7-8000.app.github.dev/exhibits/create/

Technical choices that contributed to the failure

Organizational factors

Organisational Stuff

Organizational and governance factors

Timeline

Timeline Stuff

Timeline of failure and aftermath

Create exhibit

Cancel

Create Exhibit

Section: Assessment Info

urban-waffle-4j549wnr45vcpg7-8000.app.github.dev/exhibits/create/

Image

Choose file No file chosen

Image representing this exhibit

Image reference

Image reference stuff

Reference/attribution for the image

Supporting artefacts

[Artefact]

List of supporting artefact links

Enter a valid JSON

Data issues

Data Issue Stuff

Data-related issues that contributed to the failure

New Exhibit

Section: Assessment Info

urban-waffle-4j549wnr45vcpg7-8000.app.github.dev/exhibits/33/

AI Ethics & Failures

chase Add exhibit Log out

Back to home New Domain

New Exhibit

Bookmark Edit Manage quiz Delete

Deployment Context

Deployment Context Stuff

Intended Use

Intended Use Stuff

System Type

System Type Stuff

Inputs and Assumptions

Input Stuff

Outputs Presented to Users

Output Stuff

What Went Wrong

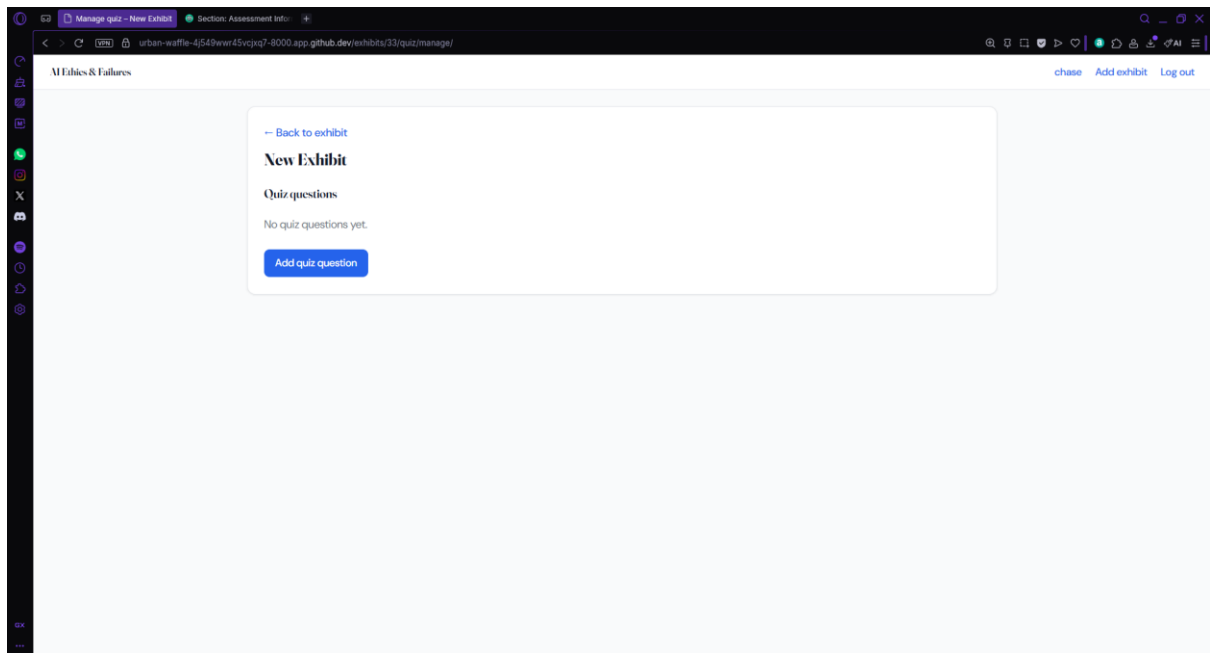
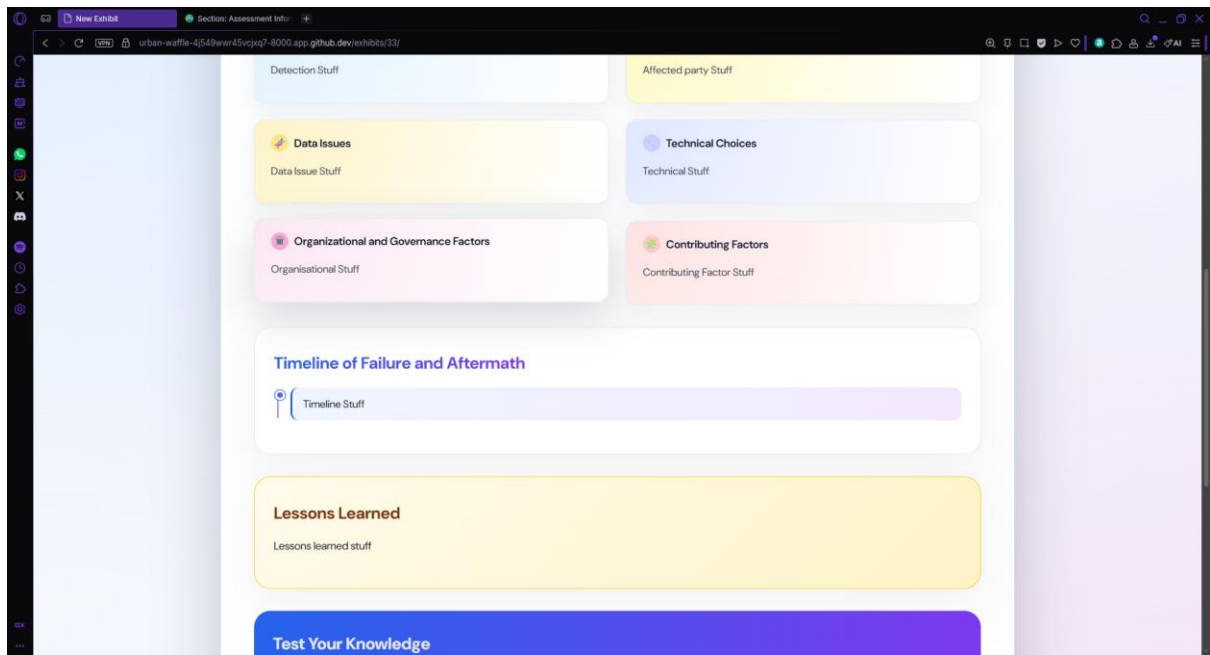
Description Stuff

How the Failure Was Detected

Detection Stuff

Who Was Affected

Affected party Stuff



Add quiz question - NewSection: Assessment info

urban-waffle-4j549wnr45vcjq7-8000.app.github.dev/exhibits/33/quiz/add/

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Add quiz question

Question

Explanation

Options (one per line)

Enter each option on a new line. The first line is the correct answer by default.

Add quiz question - NewSection: Assessment info

urban-waffle-4j549wnr45vcjq7-8000.app.github.dev/exhibits/33/quiz/add/

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Add quiz question

Question

New Question

Explanation

Because it is

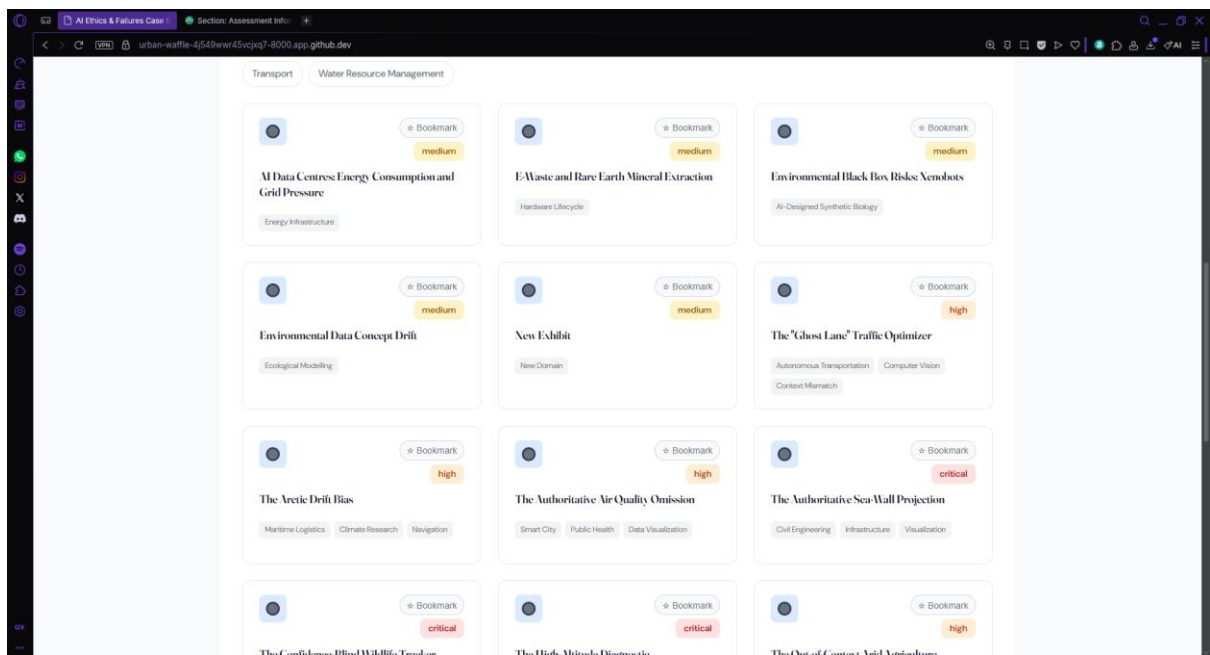
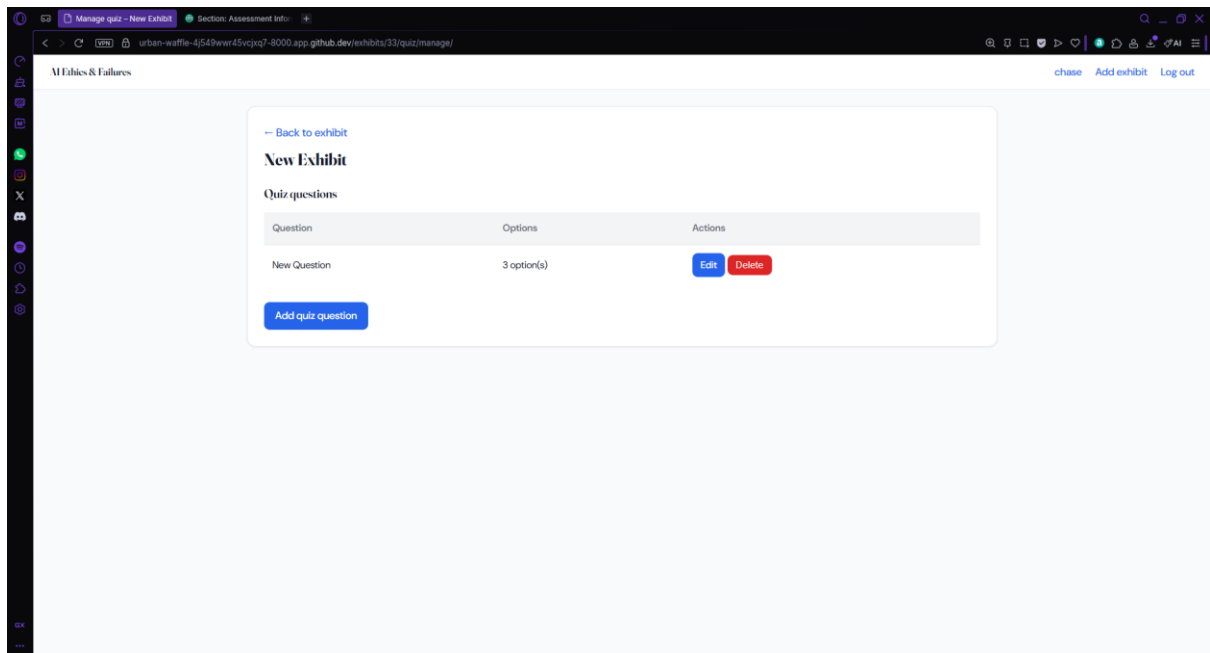
Options (one per line)

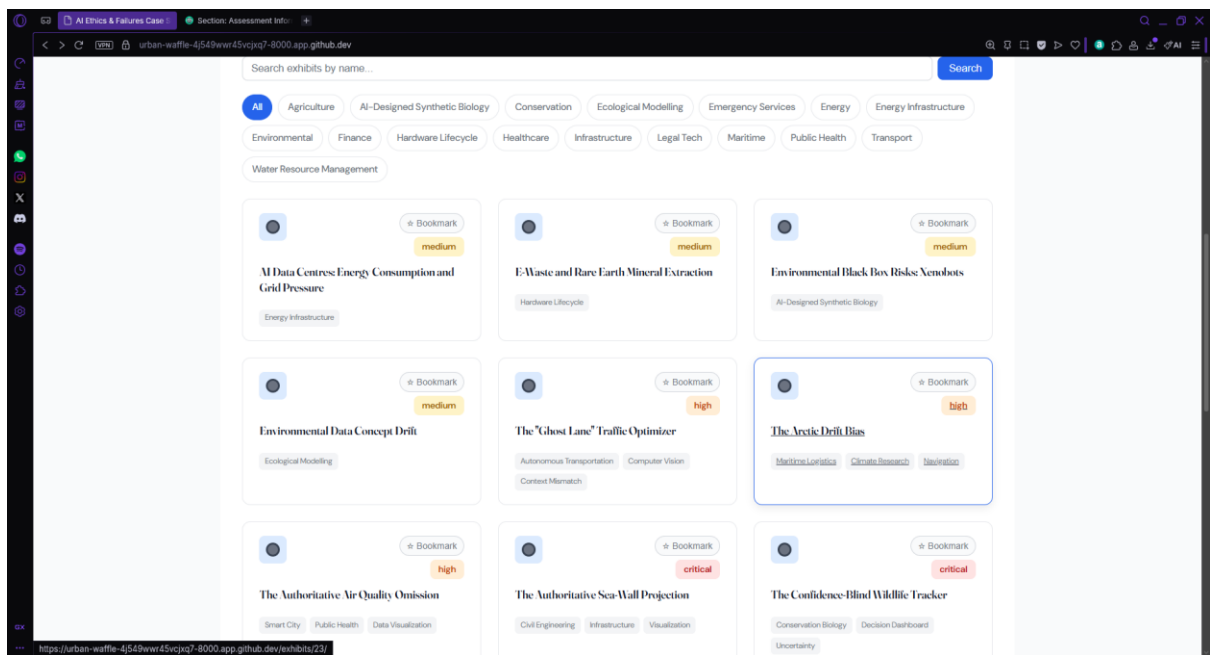
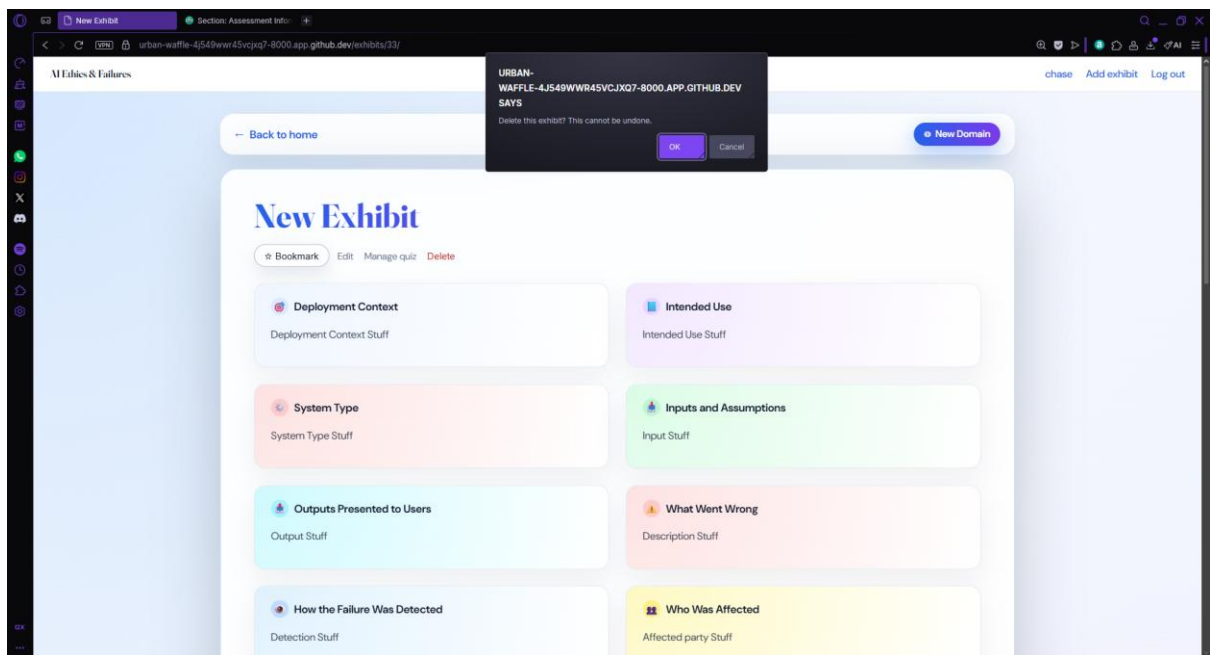
A
B
C

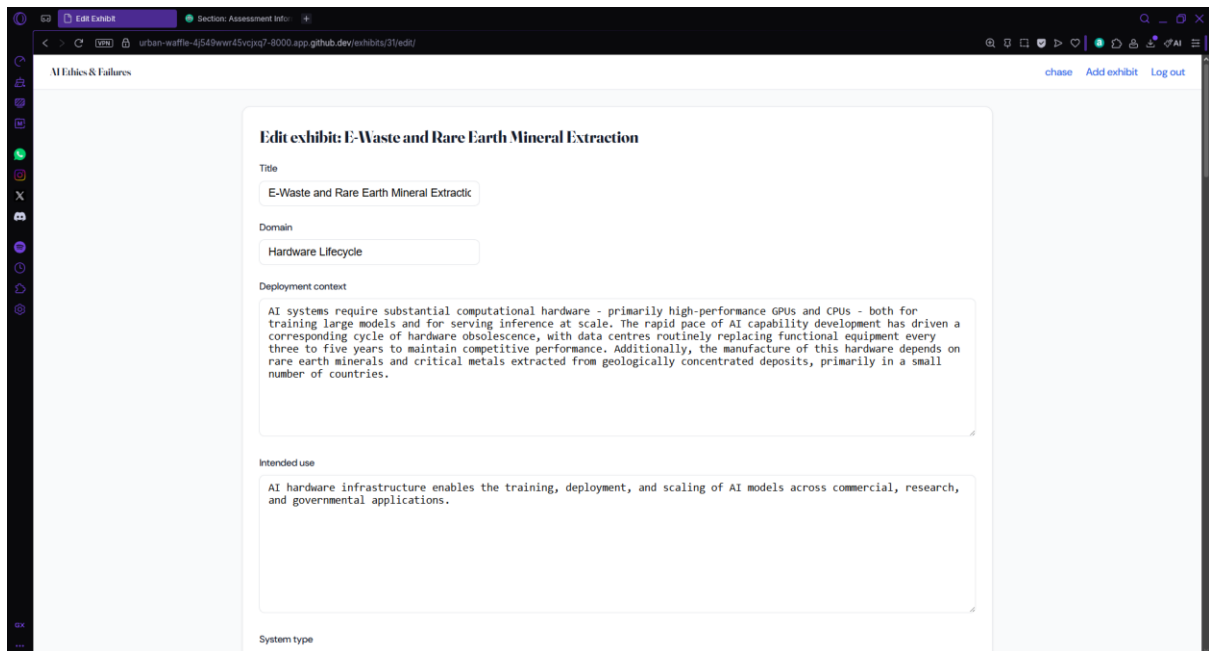
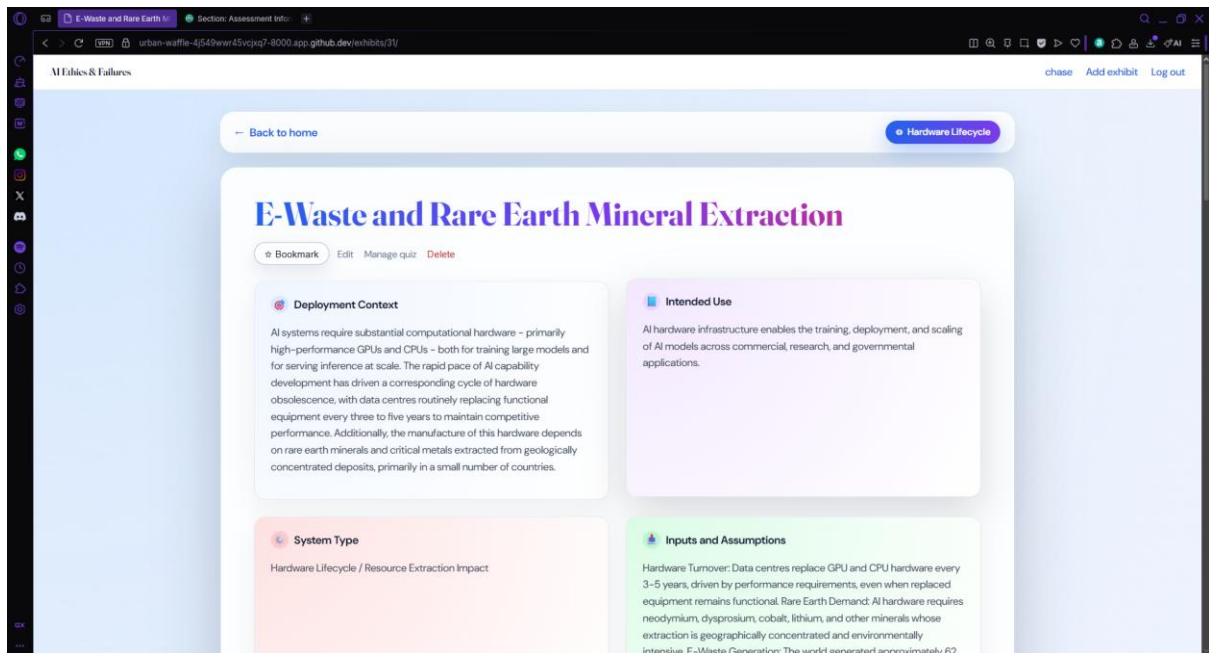
Enter each option on a new line. The first line is the correct answer by default.

Add

Cancel







AI Ethics & Failures

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Edit exhibit: E-Waste and Rare Earth Mineral Extraction

Title

E-Waste and Rare Earth Mineral Extractio

Domain

Changed Domain

Deployment context

AI systems require substantial computational hardware - primarily high-performance GPUs and CPUs - both for training large models and for serving inference at scale. The rapid pace of AI capability development has driven a corresponding cycle of hardware obsolescence, with data centres routinely replacing functional equipment every three to five years to maintain competitive performance. Additionally, the manufacture of this hardware depends on rare earth minerals and critical metals extracted from geologically concentrated deposits, primarily in a small number of countries.

Intended use

AI hardware infrastructure enables the training, deployment, and scaling of AI models across commercial, research, and governmental applications.

System type

AI Ethics & Failures Case

Section: Assessment Info

Case studies

Quizzes

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AllAgricultureAI-Designed Synthetic BiologyChanged DomainConservationEcological ModellingEmergency ServicesEnergyEnergy InfrastructureEnvironmentalFinanceHealthcareInfrastructureLegal TechMaritimePublic HealthTransportWater Resource Management

Bookmark

medium

E-Waste and Rare Earth Mineral Extraction

Changed Domain

20

Total cases

7

Critical issues

17

Categories

Admin

